DIGITAL ASSIGNMENT 3 — REVIEW 3

FinEdge

Personal Finance Management Web Application
Evidence of Testing, Version Management, System Building and Developed Functionality

Course	Software Engineering
Author	vehladkid (GitHub)
Repository	https://github.com/vehladkid/FInEdge
Release	v1.0.0

About this document

Every figure is a screenshot of the running application or of the unedited output of a command run on the project at release **v1.0.0**. Terminal figures show the first line (the command) and, for long outputs, the relevant window; the complete outputs are in `docs/review3/evidence/logs/` and can be regenerated with `docs/review3/tools/run-evidence.sh`. Application figures are captured by `frontend/tests/screenshots/capture.spec.js` using Playwright against the real stack (React → Express → PostgreSQL) with a seeded demo user.

GitHub-hosted evidence. The repository `https://github.com/vehladkid/FInEdge` was pushed after the local validation run; the repository, branches, tags, commit history, and the GitHub Actions run (CI #1, all four jobs green) are shown in Sections 4 and 5 as screenshots of the public pages. No pull requests were opened — the branches were merged locally with `--no-ff` — so there are no pull-request screenshots, and none have been fabricated.

1. 1. Integration testing
2. 2. Regression testing
3. 3. Mutation testing
4. 4. Version management
5. 5. System building
6. 6. Developed functionality — desktop (1440×900)
7. 6b. Developed functionality — mobile (390×844)
8. 7. Tools and technologies

1. Integration testing

Figure 1 — Backend integration tests

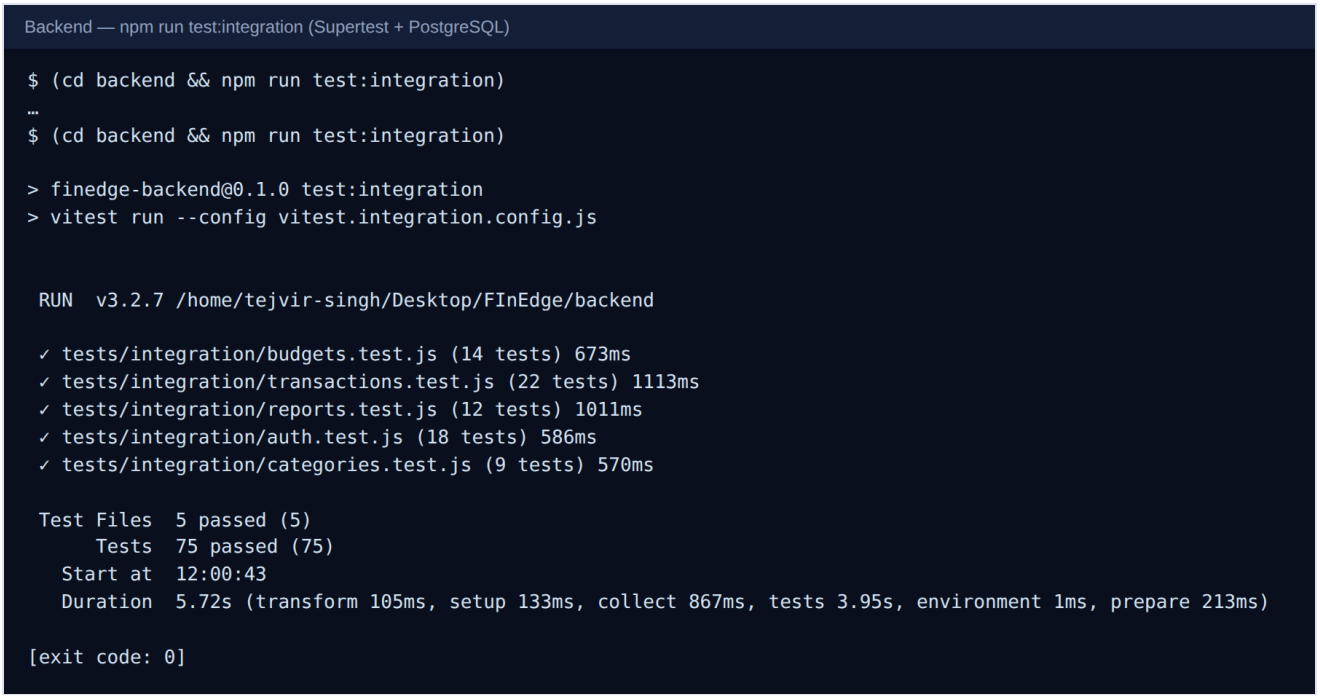


Fig. 1. npm run test:integration — 75 Supertest tests against the real Express app and a PostgreSQL *_test database. *Covers auth, transactions, categories, budgets, dashboard, analytics and insights, including cross-user isolation and concurrent registration.*

Figure 2 — Backend integration coverage

Backend — integration coverage (V8)

```
$ (cd backend && npx vitest run --config vitest.integration.config.js --coverage)
…
 % Coverage report from v8
-------------------|---------|----------|---------|---------|-------------------
File               | % Stmts | % Branch | % Funcs | % Lines | Uncovered Line #s
-------------------|---------|----------|---------|---------|-------------------
All files          |   90.94 |    90.28 |   99.08 |   90.94 |
 src               |     100 |      100 |     100 |     100 |
  app.js           |     100 |      100 |     100 |     100 |
 src/config        |     100 |        0 |       0 |     100 |
  index.js         |     100 |        0 |       0 |     100 | 5-11
 src/controllers   |     100 |      100 |     100 |     100 |
  auth.js          |     100 |      100 |     100 |     100 |
  budgets.js       |     100 |      100 |     100 |     100 |
  categories.js    |     100 |      100 |     100 |     100 |
  reports.js       |     100 |      100 |     100 |     100 |
  transactions.js  |     100 |      100 |     100 |     100 |
 src/db            |     100 |      100 |     100 |     100 |
  pool.js          |     100 |      100 |     100 |     100 |
 src/errors        |     100 |      100 |     100 |     100 |
  index.js         |     100 |      100 |     100 |     100 |
 src/middleware    |      86 |    93.33 |     100 |      86 |
  auth.js          |     100 |      100 |     100 |     100 |
  errorHandler.js  |   73.07 |    83.33 |     100 |   73.07 | 21-27
  validate.js      |     100 |      100 |     100 |     100 |
 src/repositories  |     100 |    93.47 |     100 |     100 |
  analytics.js     |     100 |      100 |     100 |     100 |
  budgets.js       |     100 |       90 |     100 |     100 | 47
  categories.js    |     100 |      100 |     100 |     100 |
  transactions.js  |     100 |    91.66 |     100 |     100 | 49
  users.js         |     100 |    83.33 |     100 |     100 | 19
 src/routes        |     100 |      100 |     100 |     100 |
  api.js           |     100 |      100 |     100 |     100 |
  health.js        |     100 |      100 |     100 |     100 |
 src/services      |   83.77 |    89.14 |     100 |   83.77 |
  aiClient.js      |      35 |    33.33 |     100 |      35 | 8-30,37-39
  analyticsCalc.js |     100 |       95 |     100 |     100 | 44
  ...icsService.js |     100 |      100 |     100 |     100 |
  authService.js   |      92 |    94.44 |     100 |      92 | 33-36
  budgetCalc.js    |     100 |       80 |     100 |     100 | 9
  budgetService.js |   95.74 |    95.23 |     100 |   95.74 | 21-22
  ...oryService.js |   95.23 |    94.73 |     100 |   95.23 | 26-27
  insightRules.js  |      61 |    61.11 |     100 |      61 | ...51,67-75,84-97
  ...ghtService.js |     100 |      100 |     100 |     100 |
  ...ionService.js |     100 |      100 |     100 |     100 |
 src/utils         |   82.89 |    80.95 |     100 |   82.89 |
  clock.js         |     100 |      100 |     100 |     100 |
  dates.js         |   77.19 |     92.3 |     100 |   77.19 | 53-65
  money.js         |     100 |      100 |     100 |     100 |
  password.js      |     100 |       50 |     100 |     100 | 8,12-13
 src/validators    |     100 |     92.3 |     100 |     100 |
  index.js         |     100 |     92.3 |     100 |     100 | 8
-------------------|---------|----------|---------|---------|-------------------
[exit code: 0]
```

Fig. 2. V8 coverage of backend/src while the integration suite runs. *All routes, controllers, repositories and middleware are exercised; gaps are the AI network call and the production error branch (both unit-tested).*

Figure 3 — Backend unit tests

Backend — npm run test:unit

```
$ (cd backend && npm run test:unit)
…
> vitest run --config vitest.unit.config.js


 RUN  v3.2.7 /home/tejvir-singh/Desktop/FInEdge/backend

 ✓ tests/unit/money.test.js (1 test) 3ms
 ✓ tests/unit/dates.test.js (26 tests) 6ms
 ✓ tests/unit/analyticsCalc.test.js (14 tests) 7ms
 ✓ tests/unit/budgetCalc.test.js (10 tests) 10ms
 ✓ tests/unit/password.test.js (10 tests) 9ms
 ✓ tests/unit/insightRules.test.js (27 tests) 25ms
 ✓ tests/unit/aiClient.test.js (8 tests) 73ms
 ✓ tests/unit/errorHandler.test.js (7 tests) 61ms
 ✓ tests/unit/validators.test.js (32 tests) 22ms

 Test Files  9 passed (9)
      Tests  135 passed (135)
   Start at  12:00:42
   Duration  463ms (transform 270ms, setup 0ms, collect 603ms, tests 217ms, environment 2ms, prepare 1.02s)

[exit code: 0]
```

Fig. 3. npm run test:unit — 135 tests of the pure business-logic modules. *Validators, dates, password policy, budget/analytics calculations, insight rules, AI client and error handler.*

Figure 4 — Frontend unit/component tests

Frontend — npm run test:unit

```
$ (cd frontend && npm run test:unit)
…
$ (cd frontend && npm run test:unit)

> finedge-frontend@0.1.0 test:unit
> vitest run


 RUN  v3.2.7 /home/tejvir-singh/Desktop/FInEdge/frontend

 ✓ tests/unit/format.test.js (7 tests) 18ms
 ✓ tests/unit/api.test.js (10 tests) 12ms
 ✓ tests/unit/dashboard.test.jsx (5 tests) 222ms
 ✓ tests/unit/transactionForm.test.jsx (7 tests) 424ms
 ✓ tests/unit/auth.test.jsx (10 tests) 706ms

 Test Files  5 passed (5)
      Tests  39 passed (39)
   Start at  12:01:22
   Duration  1.79s (transform 229ms, setup 432ms, collect 472ms, tests 1.38s, environment 2.42s, prepare 804ms)

[exit code: 0]
```

Fig. 4. npm run test:unit in frontend — 39 Vitest + React Testing Library tests. *Currency formatting, API client, session storage, login/register flow, protected routes, transaction form, dashboard and budget pages.*

Figure 5 — Frontend coverage

Frontend — npm run test:coverage

```
$ (cd frontend && npm run test:coverage)
…
 % Coverage report from v8
-------------------|---------|----------|---------|---------|-------------------
File               | % Stmts | % Branch | % Funcs | % Lines | Uncovered Line #s
-------------------|---------|----------|---------|---------|-------------------
All files          |   95.78 |     87.5 |   49.01 |   95.78 |
 components        |   97.85 |    78.57 |   72.72 |   97.85 |
  InsightsCard.jsx |     100 |       80 |     100 |     100 | 15,20
  NavBar.jsx       |    92.5 |    22.22 |      25 |    92.5 | 21-23
  ...ctedRoute.jsx |     100 |      100 |     100 |     100 |
  ...ctionForm.jsx |     100 |    90.62 |     100 |     100 | 19,30-31
 components/ui     |     100 |       80 |     100 |     100 |
  Card.jsx         |     100 |      100 |     100 |     100 |
  States.jsx       |     100 |       75 |     100 |     100 | 9
 contexts          |   88.67 |    94.11 |   66.66 |   88.67 |
  AuthContext.jsx  |   88.67 |    94.11 |   66.66 |   88.67 | 36-38,52-54
 hooks             |     100 |      100 |     100 |     100 |
  useAsync.js      |     100 |      100 |     100 |     100 |
 services          |   94.26 |    94.59 |   18.51 |   94.26 |
  api.js           |   94.33 |      100 |   66.66 |   94.33 | 59-61
  finance.js       |     100 |      100 |       0 |     100 |
  session.js       |    87.5 |       75 |     100 |    87.5 | 12-13,33-34
 utils             |     100 |      100 |     100 |     100 |
  dates.js         |     100 |      100 |     100 |     100 |
  format.js        |     100 |      100 |     100 |     100 |
-------------------|---------|----------|---------|---------|-------------------
[exit code: 0]
```

Fig. 5. npm run test:coverage in frontend. *Pages and charts are covered by the Playwright suite; see TESTING.md for what unit coverage intentionally excludes.*

2. Regression testing

Figure 6 — Playwright regression suite

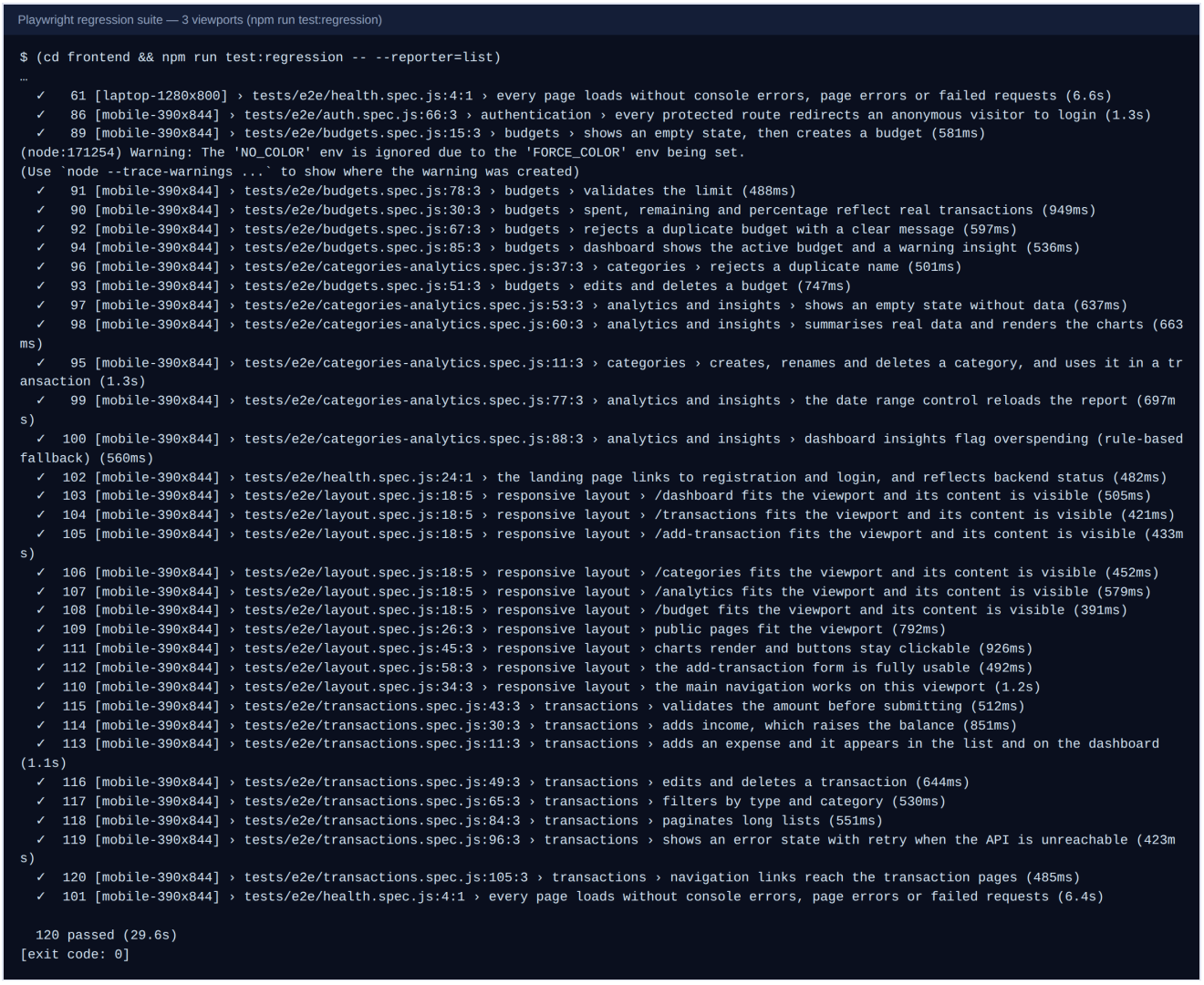


Fig. 6. npm run test:regression — 120 passed (40 tests × desktop 1440×900, laptop 1280×800, mobile 390×844). *Real browser against the real React app, API and PostgreSQL; journeys: auth, transactions, budgets, categories, analytics, layout/overflow, console health.*

Figure 7 — Mobile navigation (390×844)

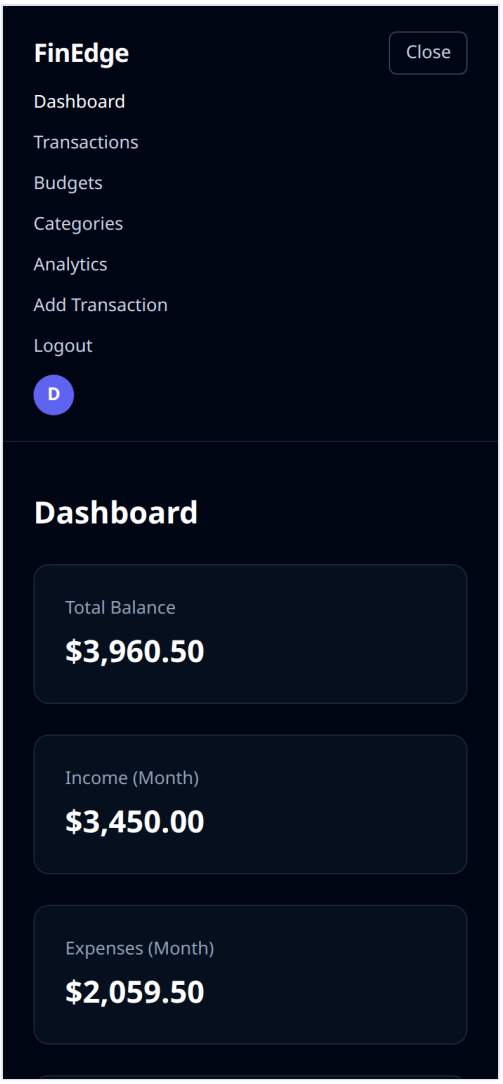


Fig. 7. The collapsible menu exercised by layout.spec.js on the mobile project. *Part of the responsive regression checks: no horizontal overflow, navigation reachable, buttons clickable.*

3. Mutation testing

Figure 8 — StrykerJS mutation report

Backend — npm run test:mutation (StrykerJS)

```
$ (cd backend && npm run test:mutation)
…
All files          |  96.53 |   96.53 |      527 |         1 |         19 |        0 |        0 |
 errors            | 100.00 |  100.00 |       11 |         0 |          0 |        0 |        0 |
  index.js         | 100.00 |  100.00 |       11 |         0 |          0 |        0 |        0 |
 middleware        | 100.00 |  100.00 |       35 |         0 |          0 |        0 |        0 |
  errorHandler.js  | 100.00 |  100.00 |       35 |         0 |          0 |        0 |        0 |
 services          |  96.94 |   96.94 |      284 |         1 |          9 |        0 |        0 |
  aiClient.js      |  91.67 |   91.67 |       44 |         0 |          4 |        0 |        0 |
  analyticsCalc.js | 100.00 |  100.00 |       70 |         1 |          0 |        0 |        0 |
  budgetCalc.js    | 100.00 |  100.00 |       23 |         0 |          0 |        0 |        0 |
  insightRules.js  |  96.71 |   96.71 |      147 |         0 |          5 |        0 |        0 |
 utils             |  95.10 |   95.10 |       97 |         0 |          5 |        0 |        0 |
  dates.js         |  92.86 |   92.86 |       65 |         0 |          5 |        0 |        0 |
  money.js         | 100.00 |  100.00 |        4 |         0 |          0 |        0 |        0 |
  password.js      | 100.00 |  100.00 |       28 |         0 |          0 |        0 |        0 |
 validators        |  95.24 |   95.24 |      100 |         0 |          5 |        0 |        0 |
  index.js         |  95.24 |   95.24 |      100 |         0 |          5 |        0 |        0 |
-------------------|--------|---------|----------|-----------|------------|----------|----------|
12:01:18 (158331) INFO MutationTestReportHelper Final mutation score of 96.53 is greater than or equal to break threshold 80
12:01:18 (158331) INFO JsonReporter Your report can be found at: file:///home/tejvir-singh/Desktop/FInEdge/backend/reports/mutation/mutation.json
12:01:18 (158331) INFO HtmlReporter Your report can be found at: file:///home/tejvir-singh/Desktop/FInEdge/backend/reports/mutation/mutation.html
12:01:18 (158331) INFO MutationTestExecutor Done in 21 seconds.
[exit code: 0]
```

Fig. 8. npm run test:mutation — mutation score 96.53 % (527 killed, 1 timeout, 19 survived of 547). *Targets 10 logic modules; survivors are analysed in TESTING.md and evidence/mutation-summary.md. The build fails below 80 %.*

4. Version management

Figure 9 — Git history graph

git log --graph --all

```
$ git log --graph --decorate --format=%h %<(10,trunc)%an %ad %d %s --date=short --all
*   709ce8a vehladkid  2026-10-04  (HEAD -> main, tag: v1.0.0) Merge fix/compose-volume-name
|\
| * 4d713f4 vehladkid  2026-10-04  (fix/compose-volume-name) fix(docker): name the PostgreSQL volume explicitly
|/
*   1810fac vehladkid  2026-10-04  Merge docs/review3-documentation: Review 3 documentation and evidence
|\
| * 08fbf56 vehladkid  2026-10-04  (docs/review3-documentation) docs: README, SRS, architecture, design of tests, diagrams and evidence for Review
3
|/
*   0d39f9a vehladkid  2026-10-04  Merge feature/mutation-hardening
|\
| * 969a370 vehladkid  2026-10-04  (feature/mutation-hardening) test(backend): assert the AI prompt's response schema
|/
*   485e287 vehladkid  2026-10-04  Merge ci/github-actions: GitHub Actions pipeline
|\
| * 1f7b5ad vehladkid  2026-10-04  (ci/github-actions) ci: lint, unit, integration, mutation, build, Playwright and Docker jobs
|/
*   b0fa7fc vehladkid  2026-10-04  Merge feature/testing: unit, integration, regression and mutation testing
|\
| * 6400ebd vehladkid  2026-10-04  (feature/testing) test(frontend): Vitest + Testing Library unit tests and Playwright regression suite
| * b96772f vehladkid  2026-10-04  test(backend): unit, integration (Supertest + PostgreSQL) and Stryker mutation tests
|/
*   ea82389 vehladkid  2026-10-04  Merge feature/frontend-integration: React app backed by the real API
|\
| * be7de90 vehladkid  2026-10-04  (feature/frontend-integration) feat(frontend): connect the UI to the API with auth, protected routes and real da
ta
|/
*   f24c99a vehladkid  2026-10-04  Merge feature/backend-api: persistent PostgreSQL-backed REST API
|\
| * 7c6be8c vehladkid  2026-10-04  (feature/backend-api) build(docker): run migrations on backend start, add postgres healthcheck and JWT/CORS conf
ig
| * 85c60eb vehladkid  2026-10-04  feat(api): authentication, transactions, categories, budgets, analytics and insights
| * da071b7 vehladkid  2026-10-04  feat(db): PostgreSQL schema, migration runner and database bootstrap script
|/
* 64e17b1 vehladkid  2026-09-11  (origin/main, origin/feature/software-design, origin/HEAD, feature/software-design) Add budget creation to the Bud
get screen
* 1f9aab3 vehladkid  2026-09-11  Remove stray .claude editor artifact from version control
* 2bf1c3f vehladkid  2026-09-10  Refactor into layered architecture and add design documentation
* b435a19 vehladkid  2026-07-31  Add core app pages (Dashboard, Analytics, Budget, Login, AddTransaction) and NavBar
* b1a1650 vehladkid  2026-07-31  Restyle landing page with Tailwind: nav, hero, status pill, feature cards
* d14cf3a vehladkid  2026-07-31  Initial FinEdge MVP skeleton: frontend, backend, docker-compose, README
```

Fig. 9. git log --graph --all at the v1.0.0 release. *GitHub Flow: feature/* and ci/* branches merged into main with --no-ff; Conventional Commit messages.*

Figure 10 — Branches

git branch -a -vv

```
$ git branch -a -vv
  ci/github-actions                     1f7b5ad ci: lint, unit, integration, mutation, build, Playwright and Docker jobs
  docs/review3-documentation            08fbf56 docs: README, SRS, architecture, design of tests, diagrams and evidence for Review 3
  feature/backend-api                   7c6be8c build(docker): run migrations on backend start, add postgres healthcheck and JWT/CORS config
  feature/frontend-integration          be7de90 feat(frontend): connect the UI to the API with auth, protected routes and real data
  feature/mutation-hardening            969a370 test(backend): assert the AI prompt's response schema
  feature/software-design               64e17b1 [origin/feature/software-design] Add budget creation to the Budget screen
  feature/testing                       6400ebd test(frontend): Vitest + Testing Library unit tests and Playwright regression suite
  fix/compose-volume-name               4d713f4 fix(docker): name the PostgreSQL volume explicitly
* main                                  709ce8a [origin/main: ahead 17] Merge fix/compose-volume-name
  remotes/origin/HEAD                   -> origin/main
  remotes/origin/feature/software-design 64e17b1 Add budget creation to the Budget screen
  remotes/origin/main                   64e17b1 Add budget creation to the Budget screen
```

Fig. 10. git branch -a -vv *Short-lived feature branches kept for traceability.*

Figure 11 — Release tag

git tag -n

```
$ git tag -n
v1.0.0          FinEdge v1.0.0 — Review 3 release
```

Fig. 11. git tag -n — v1.0.0. *Annotated tag marking the Review 3 release.*

Figure 12 — Commits and remote

git log --no-merges / git remote -v

```
$ git log --oneline --no-merges | head -30
4d713f4 fix(docker): name the PostgreSQL volume explicitly
08fbf56 docs: README, SRS, architecture, design of tests, diagrams and evidence for Review 3
969a370 test(backend): assert the AI prompt's response schema
1f7b5ad ci: lint, unit, integration, mutation, build, Playwright and Docker jobs
6400ebd test(frontend): Vitest + Testing Library unit tests and Playwright regression suite
be7de90 feat(frontend): connect the UI to the API with auth, protected routes and real data
b96772f test(backend): unit, integration (Supertest + PostgreSQL) and Stryker mutation tests
7c6be8c build(docker): run migrations on backend start, add postgres healthcheck and JWT/CORS config
85c60eb feat(api): authentication, transactions, categories, budgets, analytics and insights
da071b7 feat(db): PostgreSQL schema, migration runner and database bootstrap script
64e17b1 Add budget creation to the Budget screen
1f9aab3 Remove stray .claude editor artifact from version control
2bf1c3f Refactor into layered architecture and add design documentation
b435a19 Add core app pages (Dashboard, Analytics, Budget, Login, AddTransaction) and NavBar
b1a1650 Restyle landing page with Tailwind: nav, hero, status pill, feature cards
d14cf3a Initial FinEdge MVP skeleton: frontend, backend, docker-compose, README

$ git remote -v
origin  https://github.com/vehladkid/FInEdge.git (fetch)
origin  https://github.com/vehladkid/FInEdge.git (push)
```

Fig. 12. Non-merge commits and the configured GitHub remote. *Remote: https://github.com/vehladkid/FInEdge.git*

Figure 13 — Contributors

git shortlog -sn --all

```
$ git shortlog -sn --all
    23  vehladkid
```

Fig. 13. git shortlog -sn --all *Commit counts per author.*

Figure 14 — GitHub repository

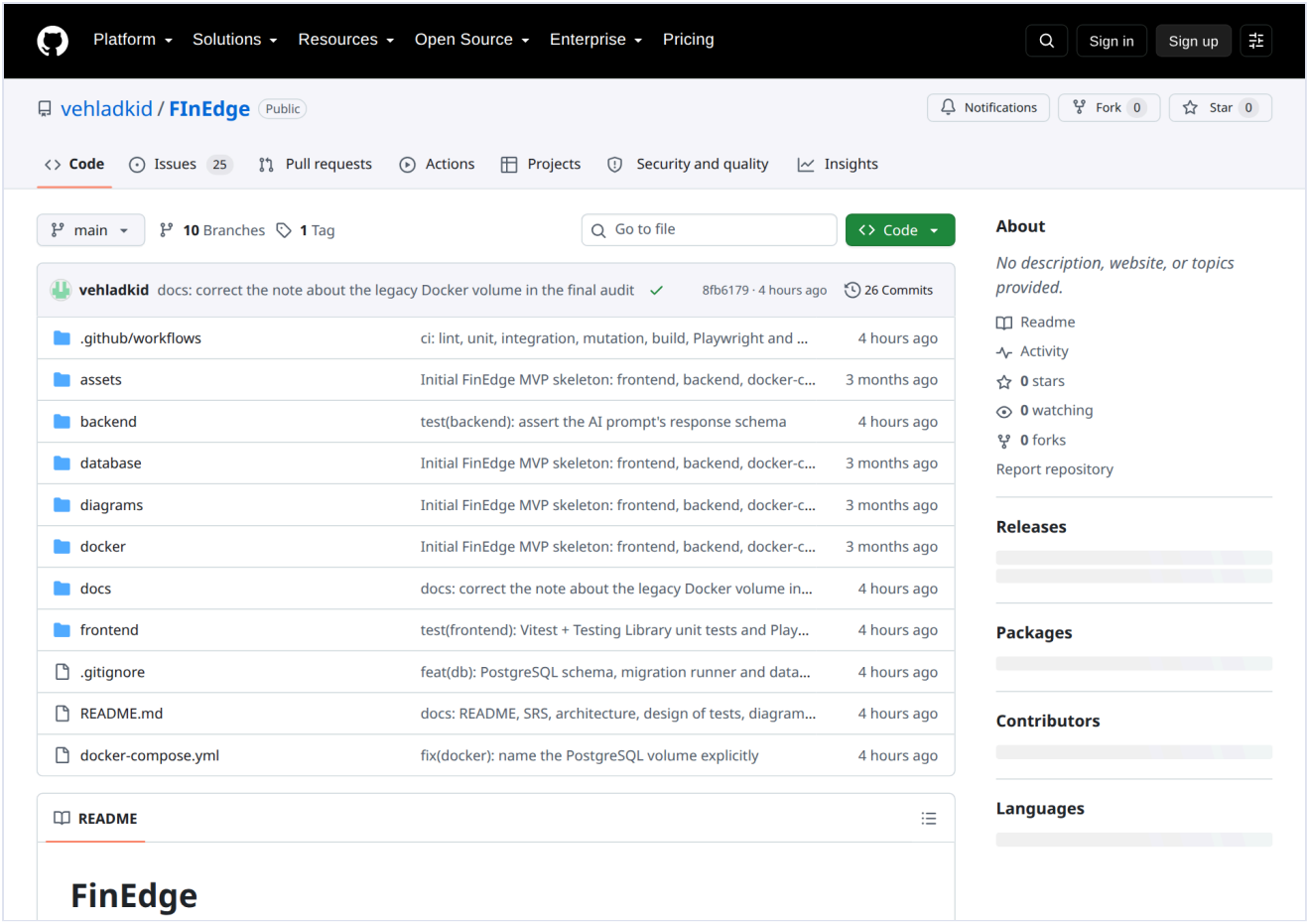


Fig. 14. https://github.com/vehladkid/FInEdge after pushing main, 8 feature/ci/docs/fix branches and the v1.0.0 tag. *Public repository; latest commit shows the passing CI check mark.*

Figure 15 — GitHub branches

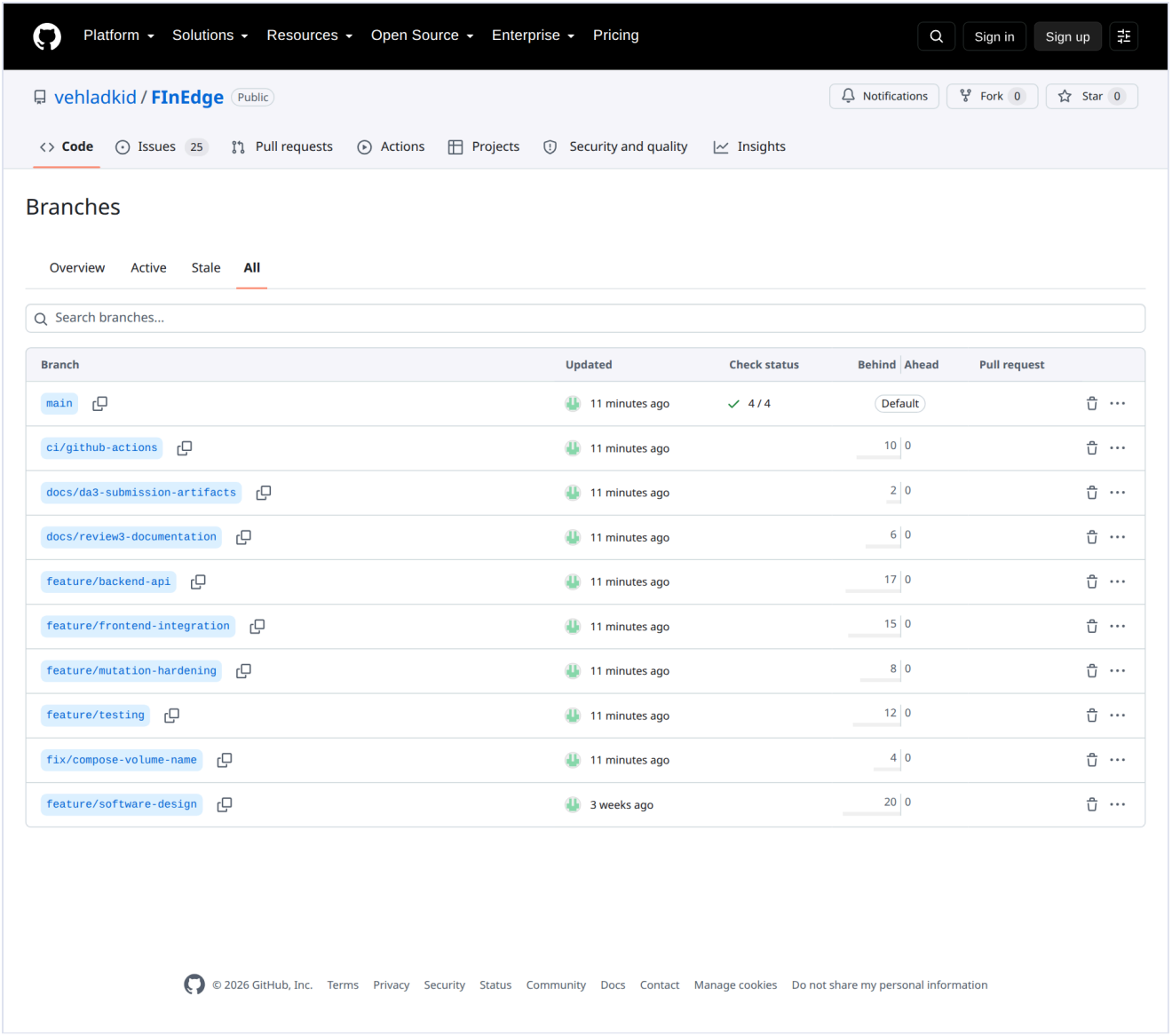


Fig. 15. All branches on GitHub. *Includes the feature/*, ci/*, docs/* and fix/* branches that were merged into main.*

Figure 16 — GitHub tags

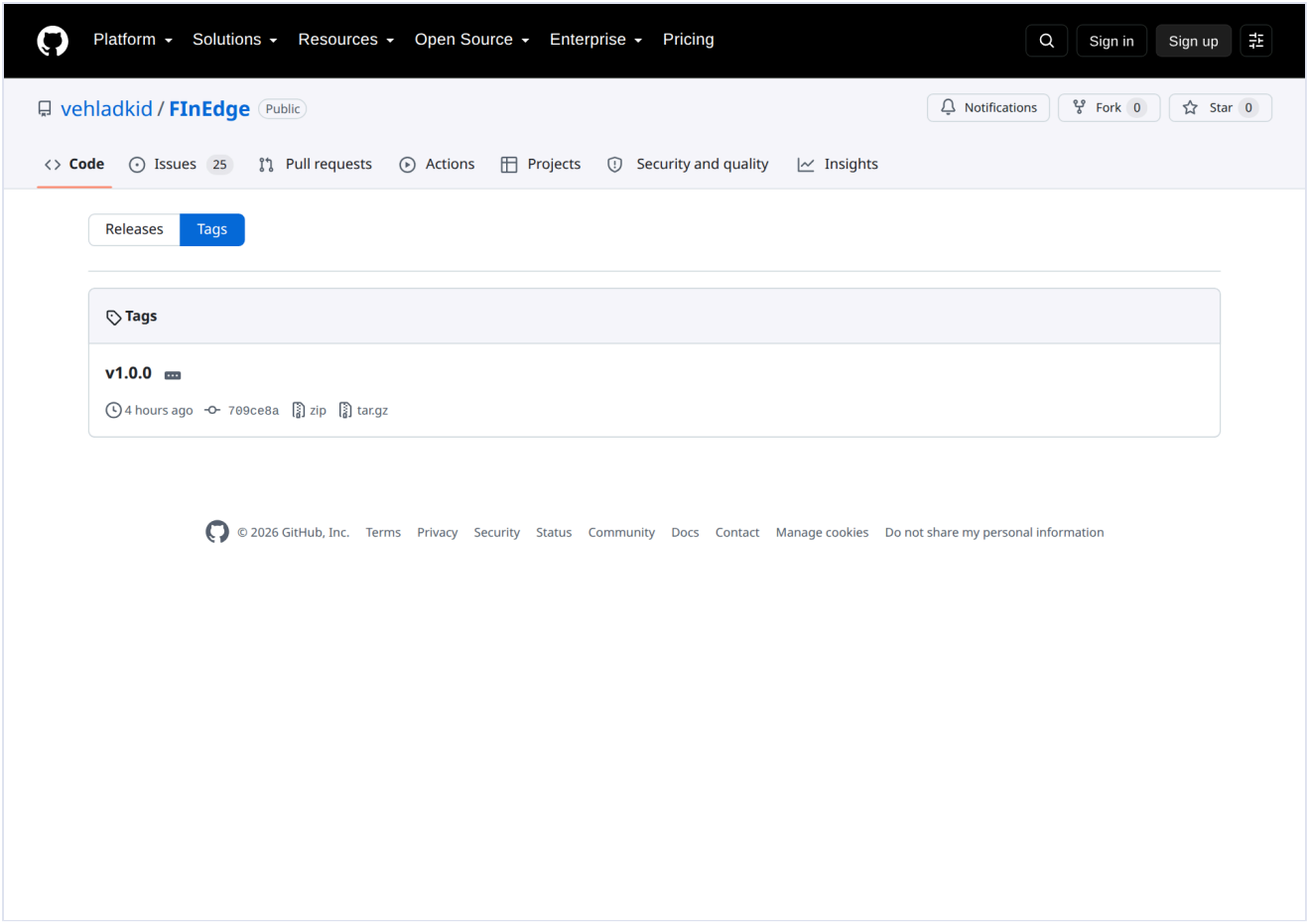


Fig. 16. Release tag v1.0.0 on GitHub.

Figure 17 — GitHub commit history

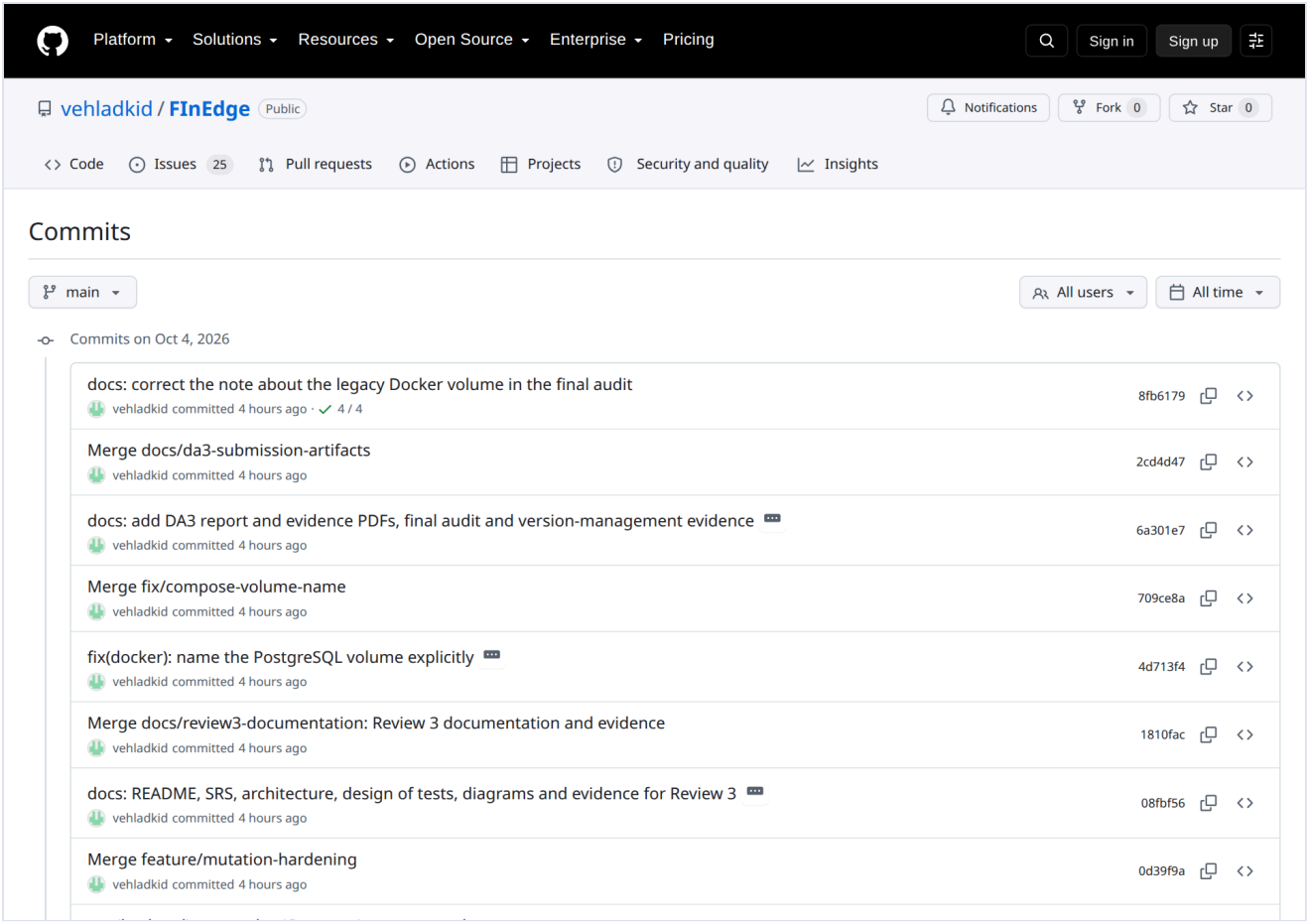


Fig. 17. Commit history of main on GitHub. *Conventional Commit messages and --no-ff merge commits.*

5. System building

Figure 18 — Production build

Frontend — npm run build

```
$ (cd frontend && npm run build)
…
> vite build

vite v5.4.21 building for production...
transforming...
✓ 652 modules transformed.
rendering chunks...
computing gzip size...
dist/index.html                   0.39 kB │ gzip:   0.26 kB
dist/assets/index-BXDBZeSy.css   14.59 kB │ gzip:   3.62 kB
dist/assets/index-D1KymGyo.js   629.43 kB │ gzip: 187.12 kB

(!) Some chunks are larger than 500 kB after minification. Consider:
- Using dynamic import() to code-split the application
- Use build.rollupOptions.output.manualChunks to improve chunking: https://rollupjs.org/configuration-options/#output-manualchunks
- Adjust chunk size limit for this warning via build.chunkSizeWarningLimit.
✓ built in 2.22s
[exit code: 0]
```

Fig. 18. npm run build (Vite) — success. *Warning about the 629 kB chunk is documented in TESTING.md.*

Figure 19 — Backend lint

Backend — npm run lint

```
$ (cd backend && npm run lint)
…
$ (cd backend && npm run lint)

> finedge-backend@0.1.0 lint
> eslint .

[exit code: 0]
```

Fig. 19. npm run lint — ESLint, no findings.

Figure 20 — Frontend lint

Frontend — npm run lint

```
$ (cd frontend && npm run lint)
…
$ (cd frontend && npm run lint)

> finedge-frontend@0.1.0 lint
> eslint .

[exit code: 0]
```

Fig. 20. npm run lint — ESLint with react and react-hooks rules, no findings.

Figure 21 — Docker build — backend

docker build ./backend

```
$ (cd . && docker build -t finedge-backend ./backend)
…
 ---> df3cf276d197
Step 5/7 : COPY . .
 ---> Using cache
 ---> 88b5d2b22afd
Step 6/7 : EXPOSE 5000
 ---> Using cache
 ---> 660c118cf6bb
Step 7/7 : CMD ["sh", "-c", "node src/db/migrate.js && node src/server.js"]
 ---> Using cache
 ---> da38df9a6477
Successfully built da38df9a6477
Successfully tagged finedge-backend:latest
[exit code: 0]
```

Fig. 21. docker build ./backend — success. *Image runs migrations then starts the API.*

Figure 22 — Docker build — frontend

docker build ./frontend

```
$ (cd . && docker build -t finedge-frontend ./frontend)
…
Step 5/7 : COPY . .
 ---> db1ef8bfbab6
Step 6/7 : EXPOSE 5173
 ---> Running in 8ee2cfebcd10
 ---> Removed intermediate container 8ee2cfebcd10
 ---> 016bb5944382
Step 7/7 : CMD ["npm", "run", "dev"]
 ---> Running in ef1414e091f9
 ---> Removed intermediate container ef1414e091f9
 ---> 5c9ecf6dcdd7
Successfully built 5c9ecf6dcdd7
Successfully tagged finedge-frontend:latest
[exit code: 0]
```

Fig. 22. docker build ./frontend — success.

Figure 23 — Running stack

docker-compose ps

```
$ docker-compose ps
       Name                     Command                State                         Ports
-------------------------------------------------------------------------------------------------------------
finedge_backend_1    docker-entrypoint.sh sh -c ...   Up             0.0.0.0:5000->5000/tcp,:::5000->5000/tcp
finedge_frontend_1   docker-entrypoint.sh npm r ...   Up             0.0.0.0:5173->5173/tcp,:::5173->5173/tcp
finedge_postgres_1   docker-entrypoint.sh postgres    Up (healthy)   0.0.0.0:5433->5432/tcp,:::5433->5432/tcp
```

Fig. 23. docker-compose ps — postgres (healthy), backend and frontend up. *Started from an empty database; migration 001_init.sql applied automatically.*

Figure 24 — CI pipeline definition

.github/workflows/ci.yml

```
$ sed -n 1,200p .github/workflows/ci.yml
name: CI

on:
  push:
    branches: [main]
  pull_request:

env:
  PGUSER: finedge
  PGPASSWORD: finedge

jobs:
  backend:
    name: Backend — lint, unit, integration, mutation
    runs-on: ubuntu-latest
    services:
      postgres:
        image: postgres:16
        env:
          POSTGRES_USER: finedge
          POSTGRES_PASSWORD: finedge
          POSTGRES_DB: finedge_test
        ports: ["5432:5432"]
        options: >-
          --health-cmd "pg_isready -U finedge -d finedge_test"
          --health-interval 5s --health-timeout 3s --health-retries 10
    env:
      TEST_DATABASE_URL: postgresql://finedge:finedge@localhost:5432/finedge_test
    defaults:
      run:
        working-directory: backend
    steps:
      - uses: actions/checkout@v4
      - uses: actions/setup-node@v4
        with:
          node-version: 22
          cache: npm
          cache-dependency-path: backend/package-lock.json
      - run: npm ci
      - run: npm run lint
      - run: npm run test:unit
      - run: npm run test:integration
      - run: npm run test:mutation
      - uses: actions/upload-artifact@v4
        if: always()
        with:
          name: backend-mutation-report
          path: backend/reports/mutation/
          if-no-files-found: ignore

  frontend:
    name: Frontend — lint, unit, build
    runs-on: ubuntu-latest
    defaults:
      run:
        working-directory: frontend
    steps:
      - uses: actions/checkout@v4
      - uses: actions/setup-node@v4
        with:
          node-version: 22
          cache: npm
          cache-dependency-path: frontend/package-lock.json
      - run: npm ci
      - run: npm run lint
      - run: npm run test:unit
      - run: npm run build

  e2e:
```

Fig. 24. .github/workflows/ci.yml — jobs backend, frontend, e2e, docker. *The run of this workflow on GitHub is shown in the two figures that follow.*

Figure 25 — GitHub Actions — workflow runs

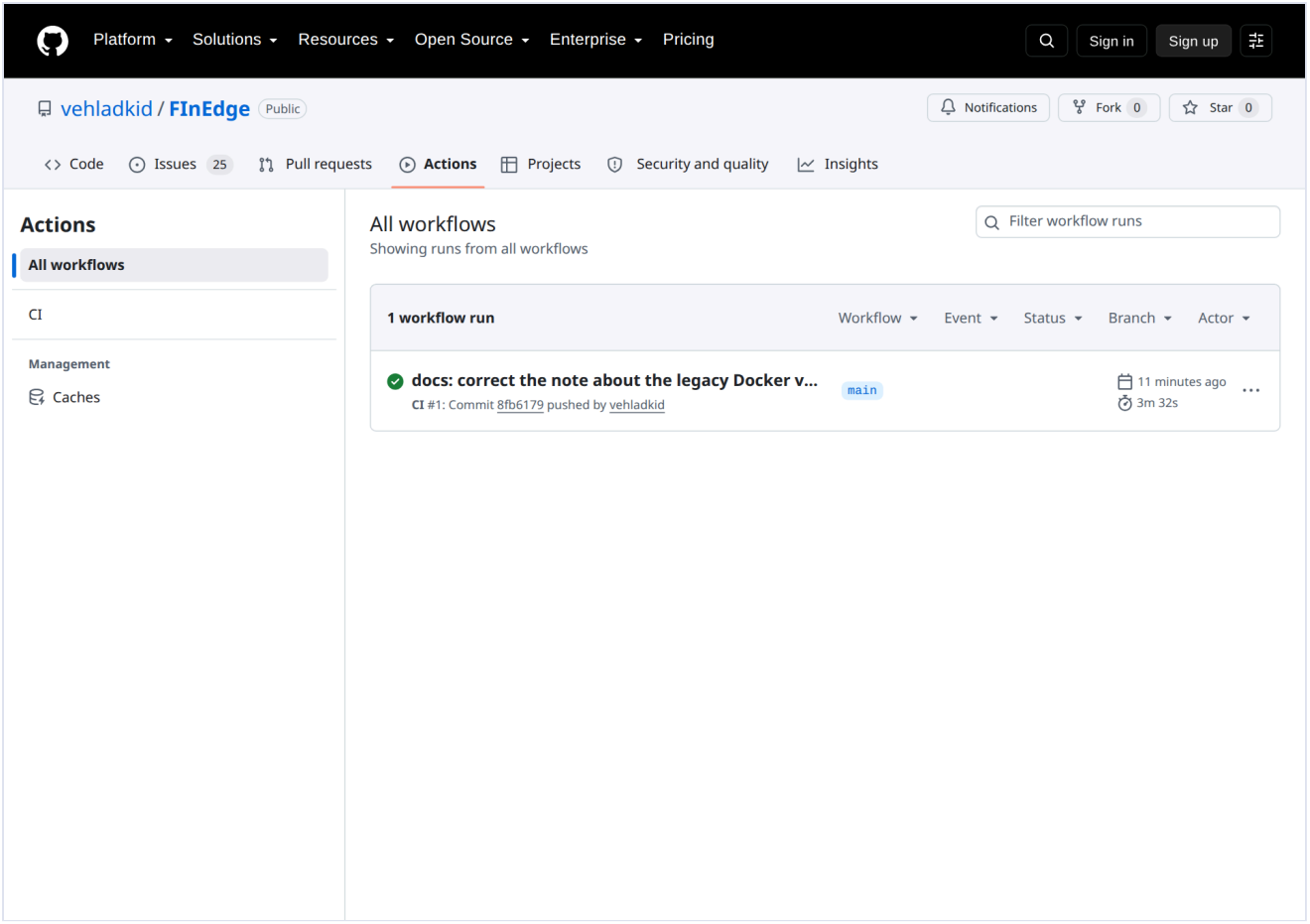


Fig. 25. CI #1 on main: success. *Triggered by the push of commit 8fb6179.*

Figure 26 — GitHub Actions — CI run #1

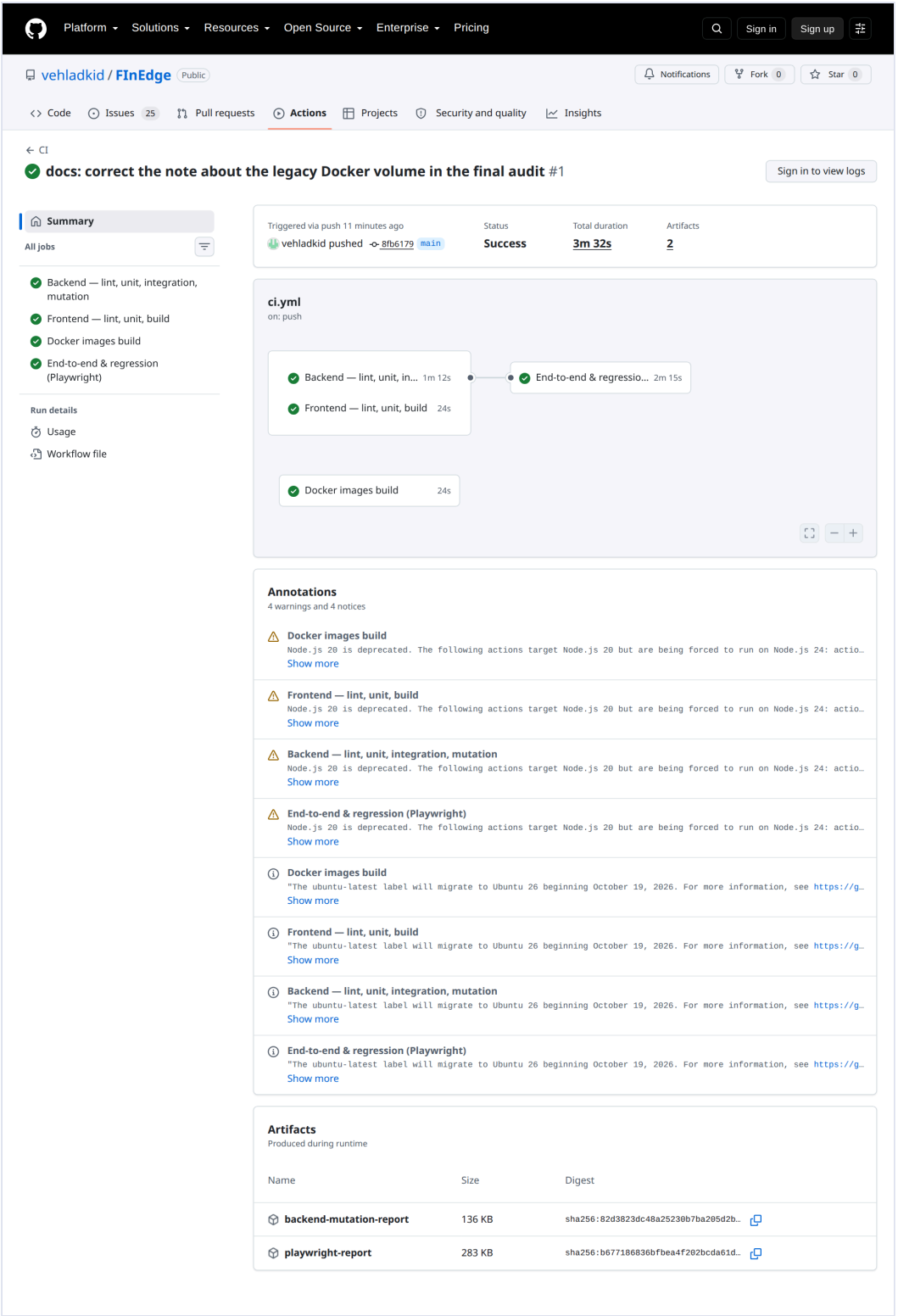


Fig. 26. All four jobs passed on a clean GitHub-hosted runner: Backend (lint, unit, integration, mutation), Frontend (lint, unit, build), End-to-end & regression (Playwright), Docker images build; 3m 32s; artifacts: mutation report and Playwright report. *The run page shows only Node-version and ubuntu-latest deprecation notices (annotations), no failures.*

6. Developed functionality — desktop (1440×900)

Figure 27 — Landing page

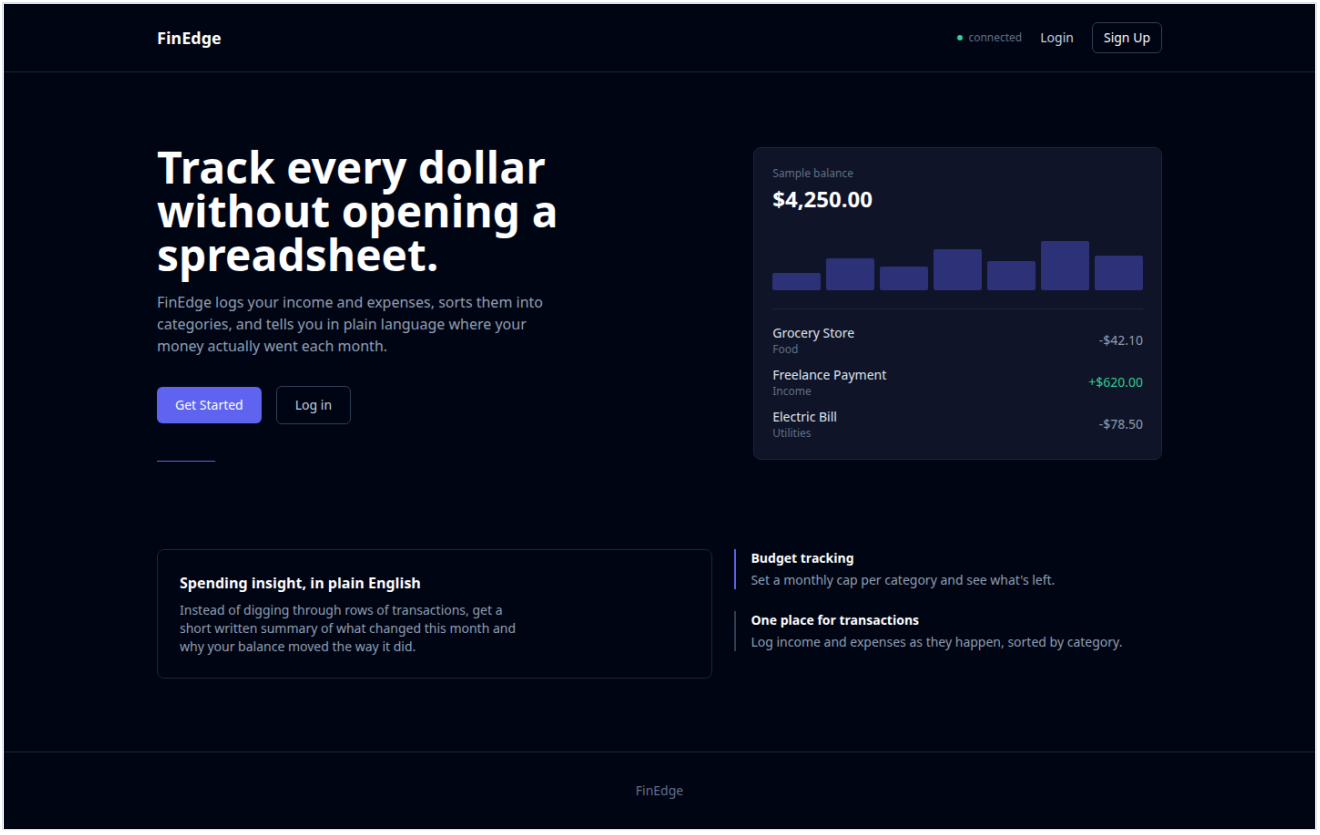


Fig. 27. Landing page; status indicator shows the backend is connected. *The preview panel is a labelled sample, not user data.*

Figure 28 — Registration

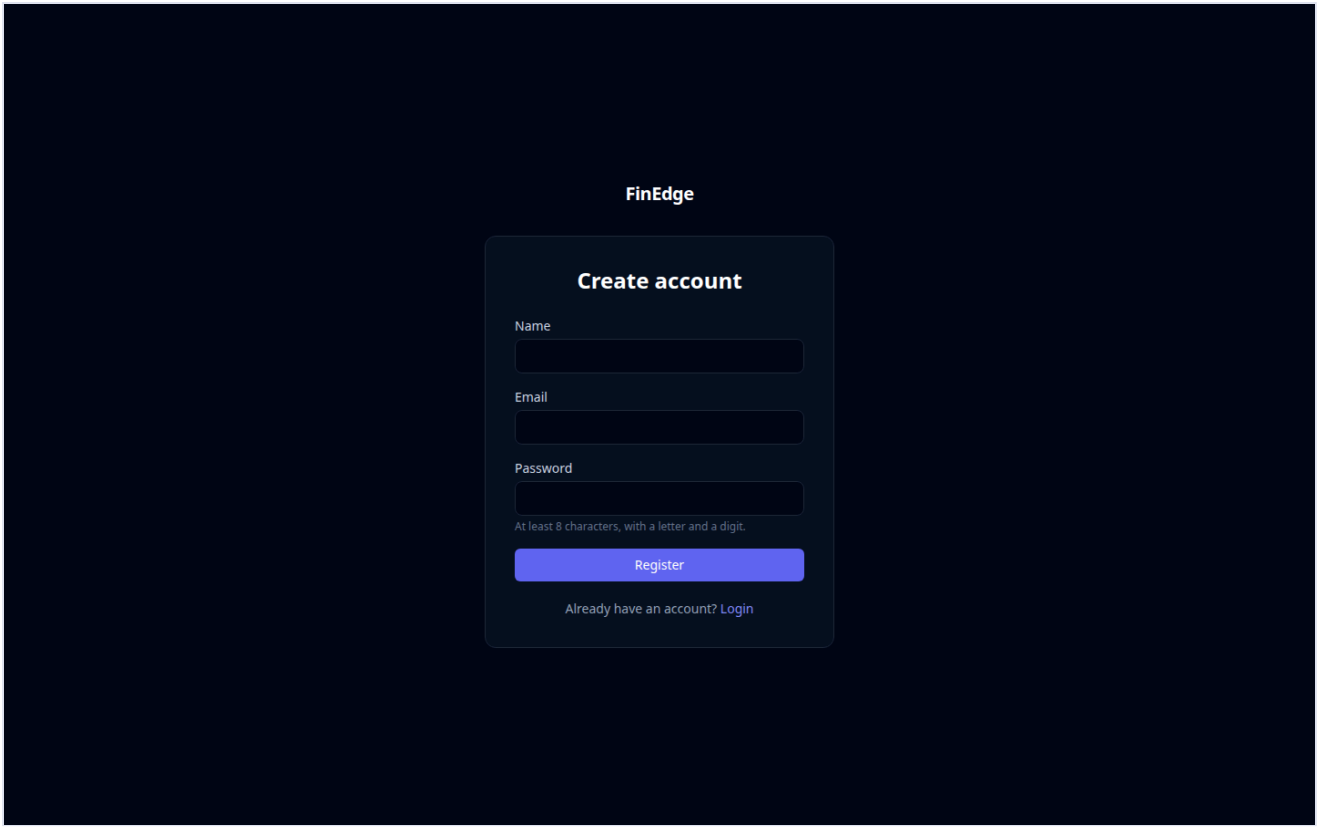


Fig. 28. Create-account form. *FR-01.*

Figure 29 — Login — invalid credentials

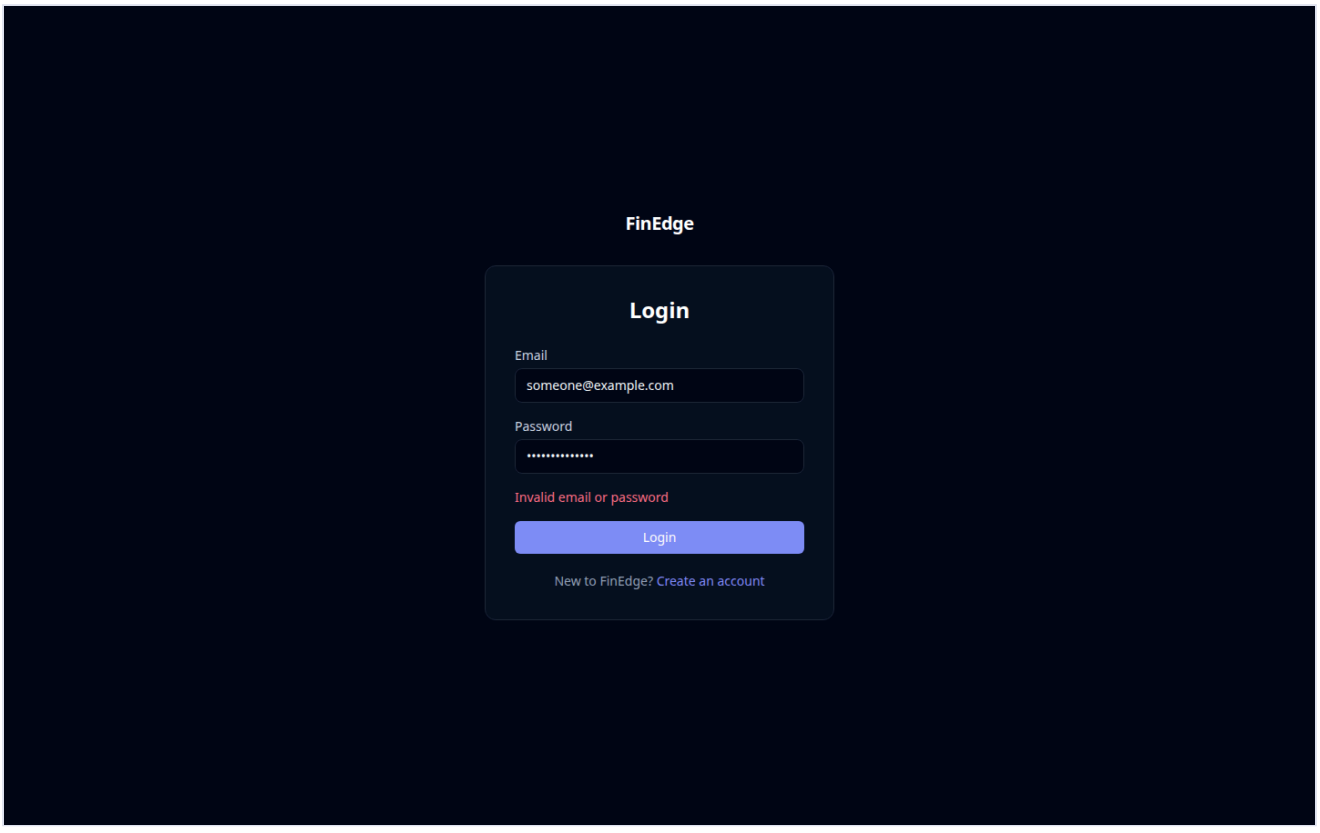


Fig. 29. Server error message shown for a wrong password. *FR-04; identical message for unknown e-mail.*

Figure 30 — Dashboard

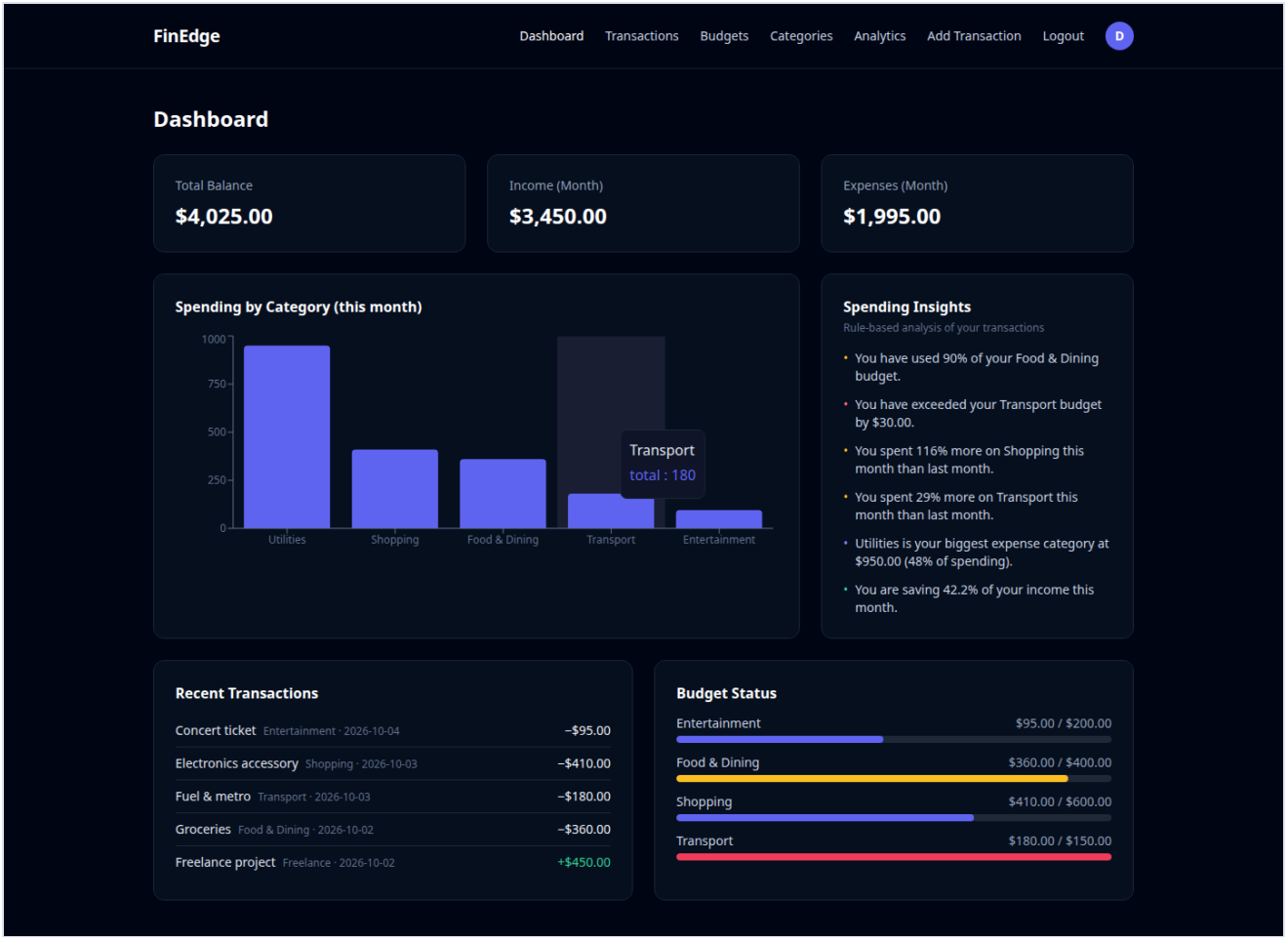


Fig. 30. Balance, month totals, spending by category, insights, recent transactions and budget status — all from the database. *FR-17, FR-19.*

Figure 31 — Transactions

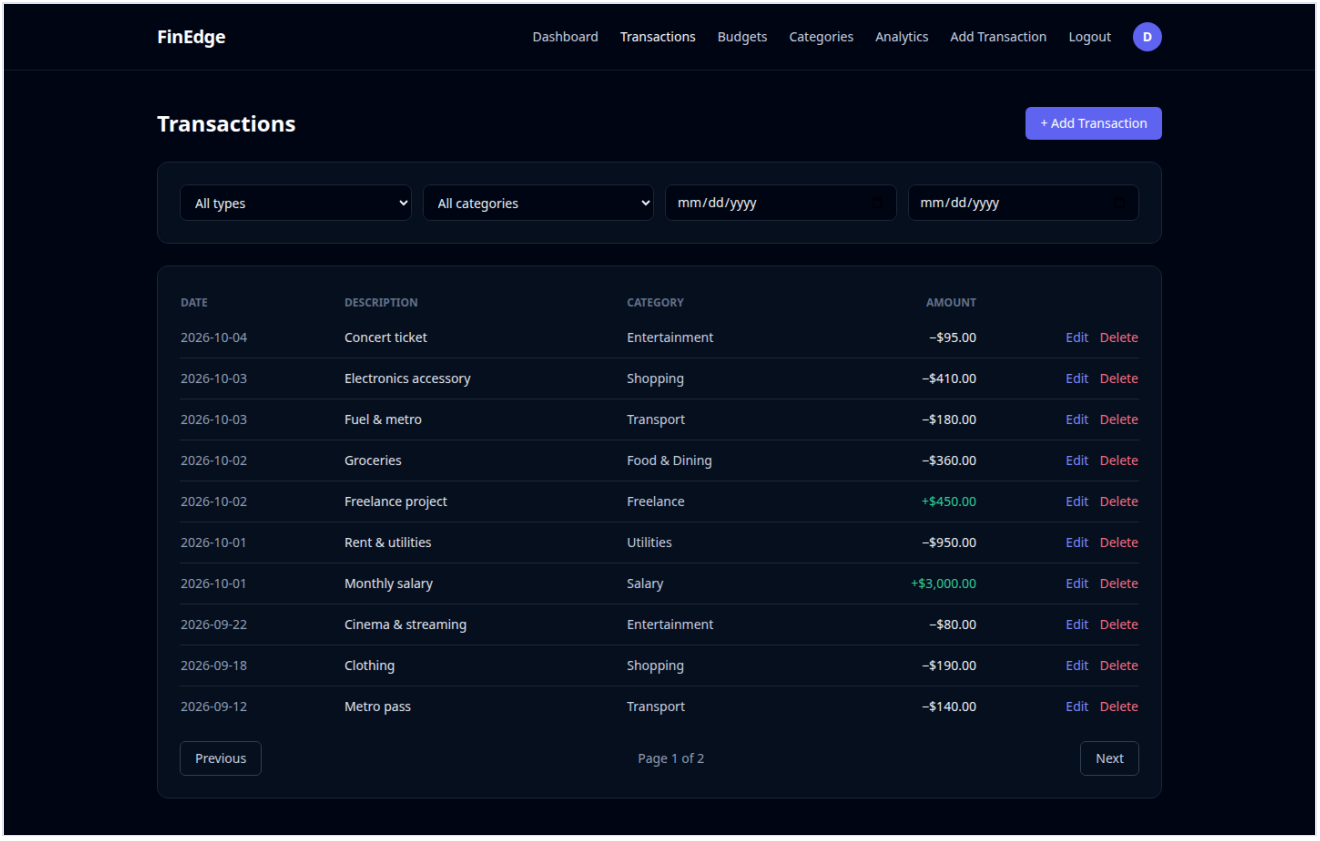


Fig. 31. Filterable, paginated transaction list with edit and delete. *FR-10, FR-11.*

Figure 32 — Add transaction — validation

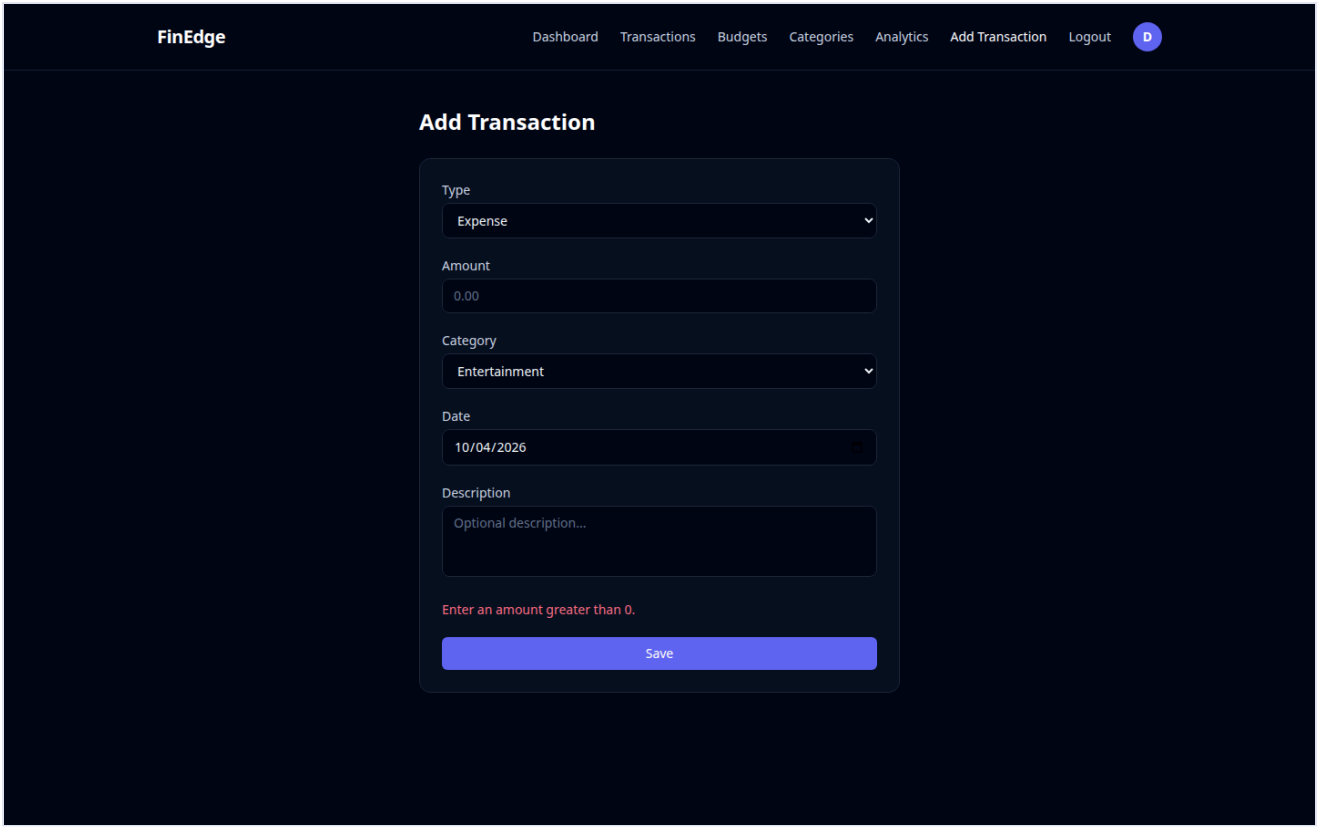


Fig. 32. Client-side validation message. *FR-08.*

Figure 33 — Add transaction — saved

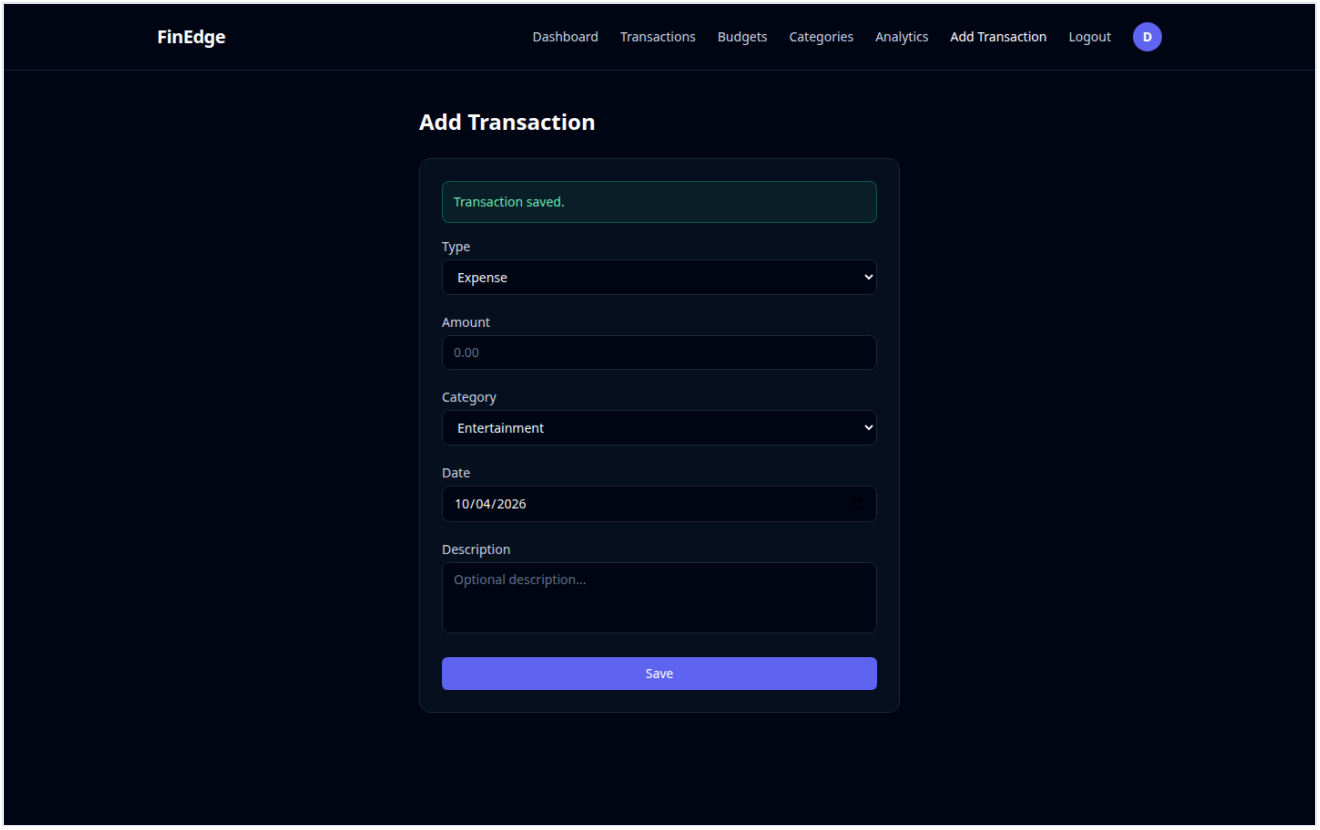


Fig. 33. Success state after saving to PostgreSQL. *FR-08, FR-09.*

Figure 34 — Edit transaction

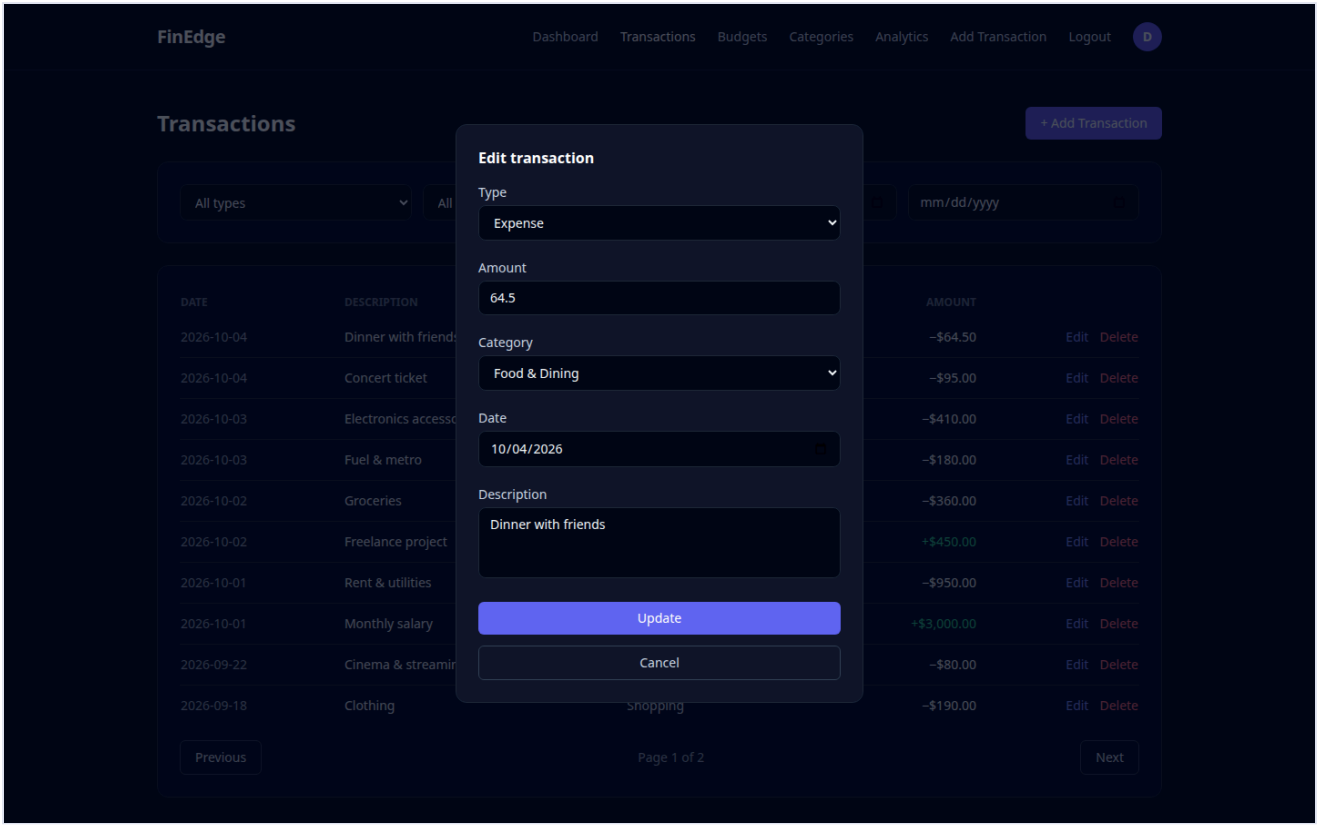


Fig. 34. Edit dialog pre-filled with the transaction. *FR-11.*

Figure 35 — Budgets

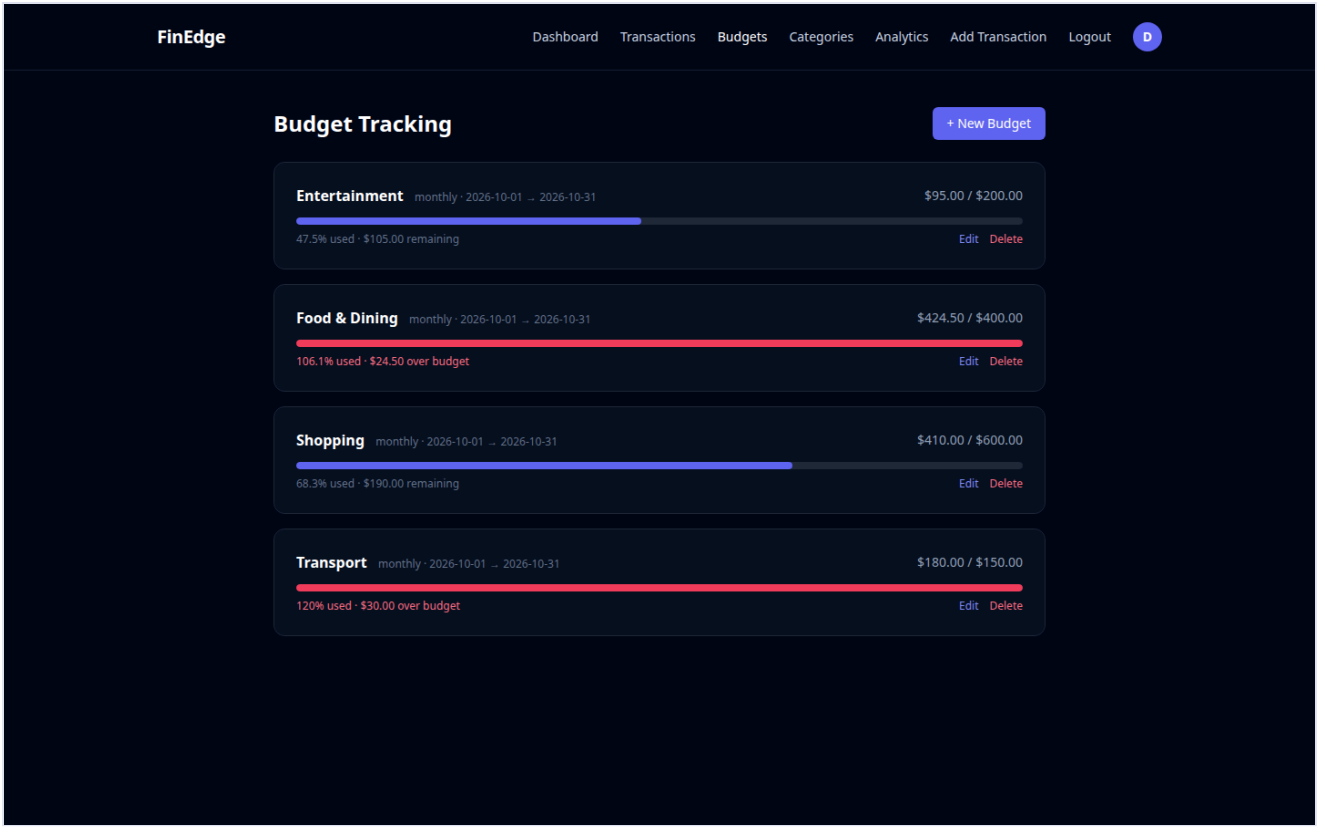


Fig. 35. Spent, remaining, percentage and status calculated by the backend; two budgets exceeded or near limit. *FR-15, FR-16.*

Figure 36 — New budget

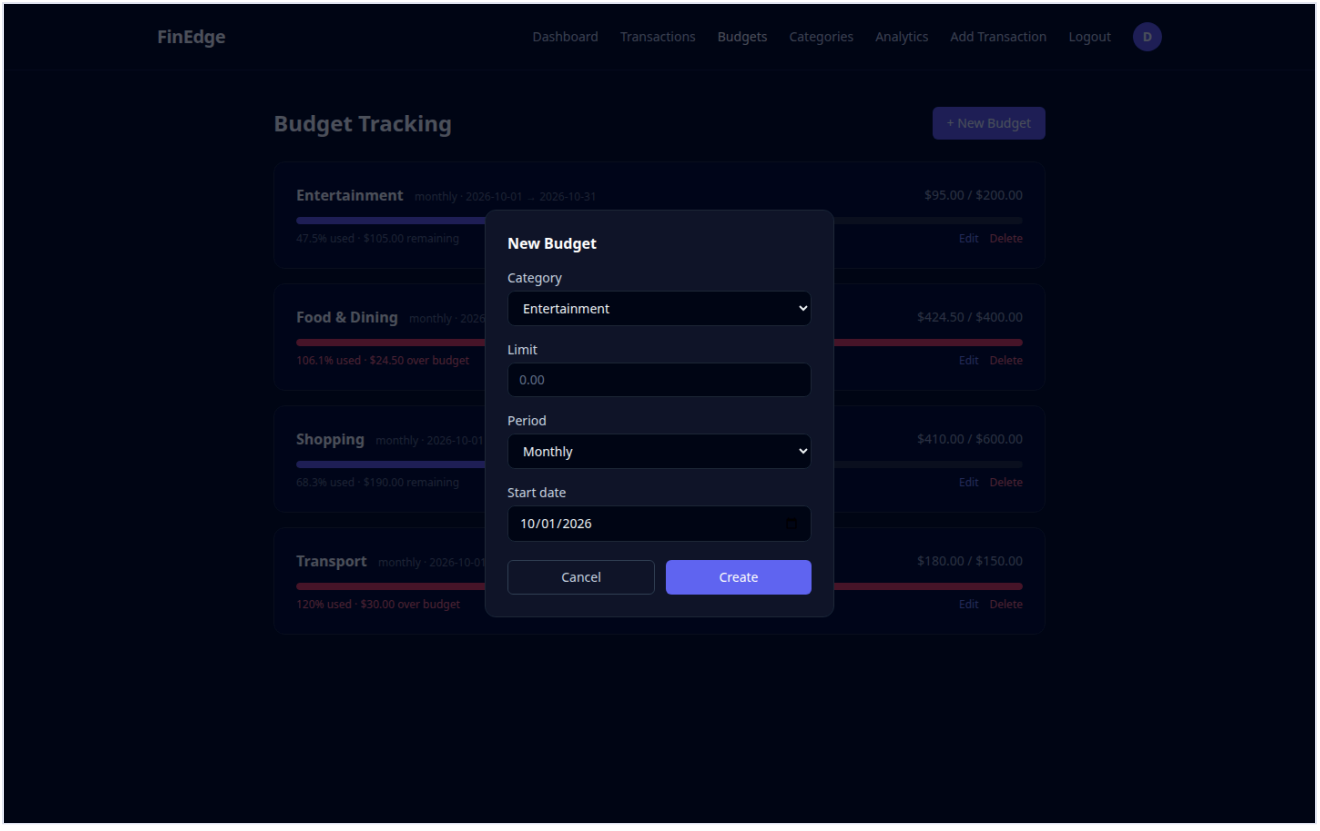


Fig. 36. Create-budget dialog (category, limit, period, start date). *FR-15.*

Figure 37 — Categories

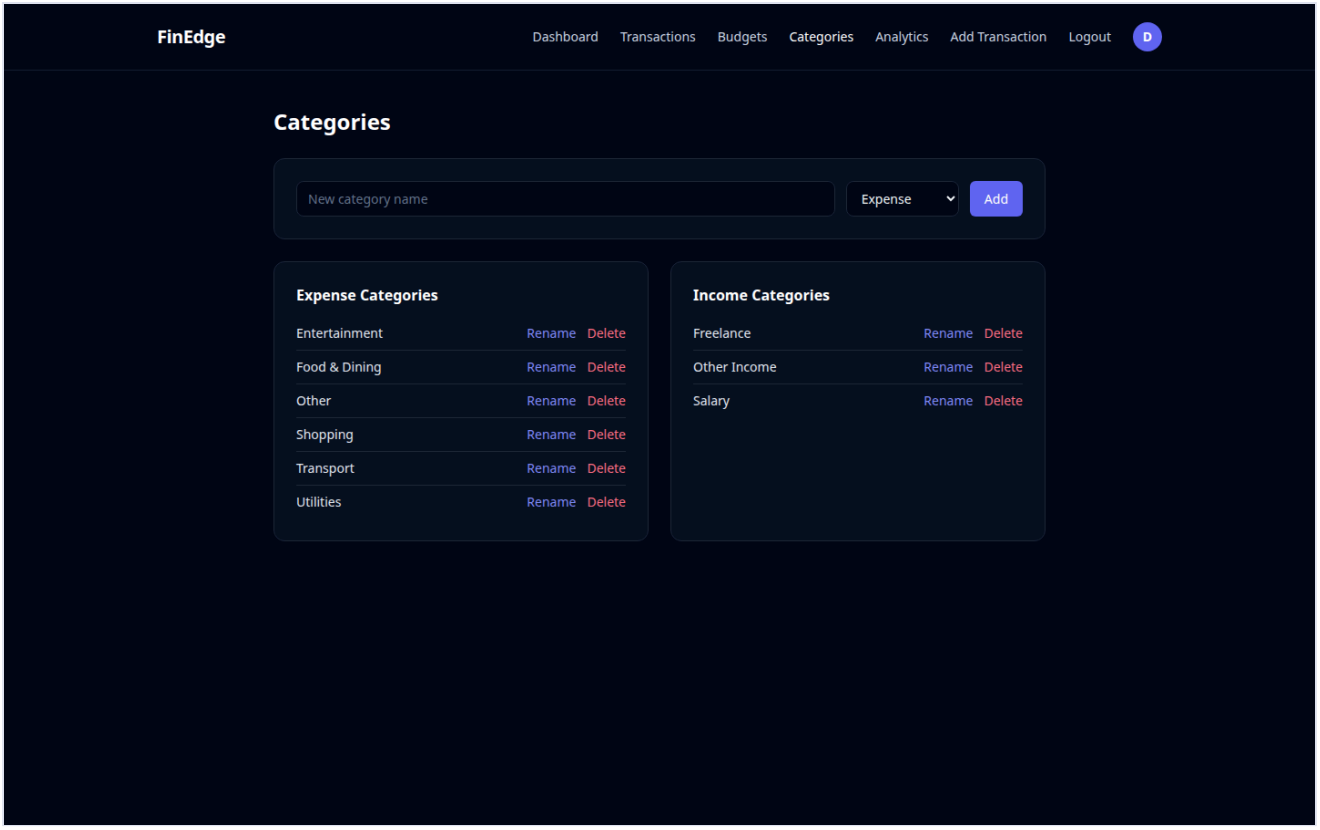


Fig. 37. Default and custom categories by type. *FR-13, FR-14.*

Figure 38 — Analytics

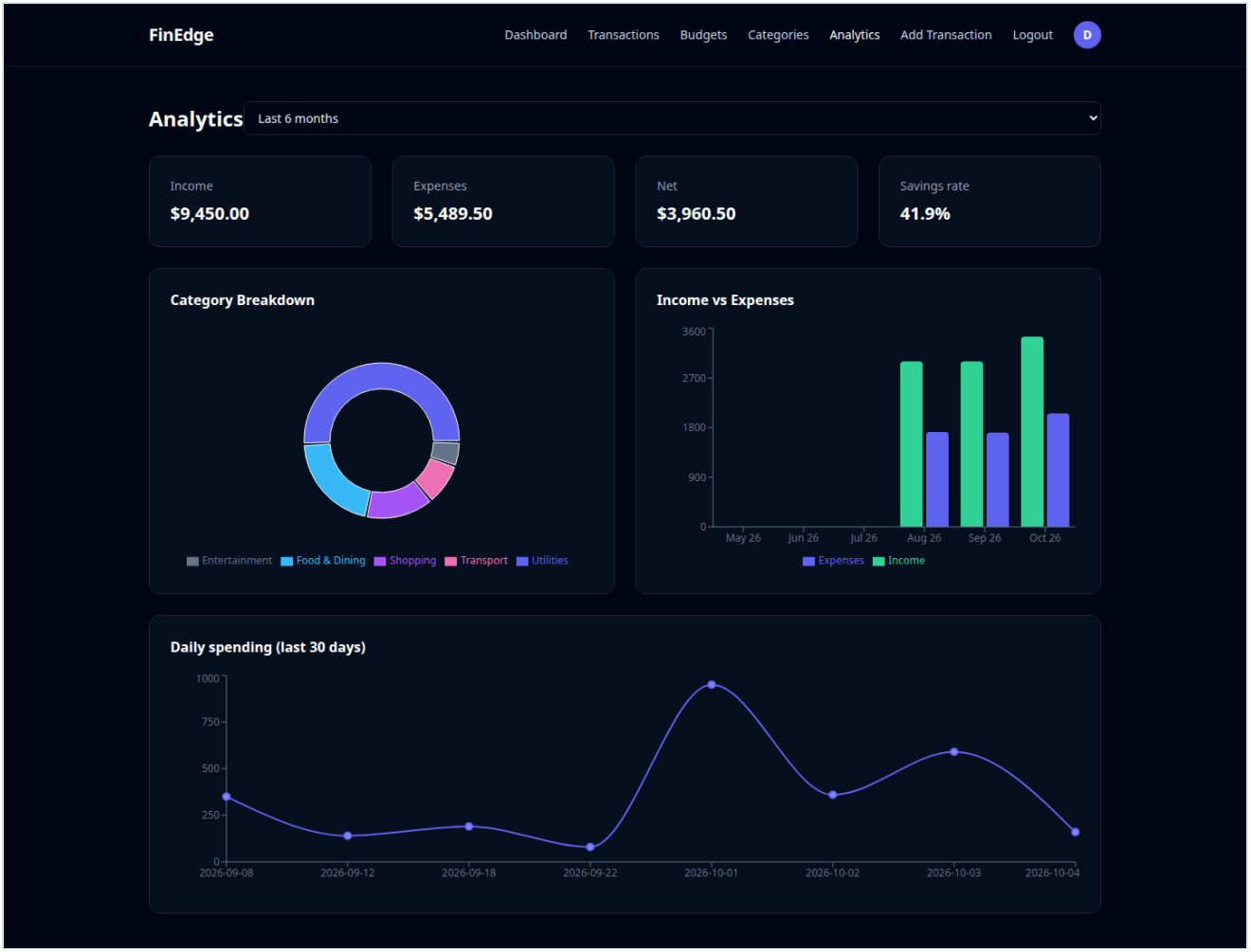


Fig. 38. Category breakdown, income vs expenses, daily spending, savings rate. *FR-18.*

Figure 39 — Insights

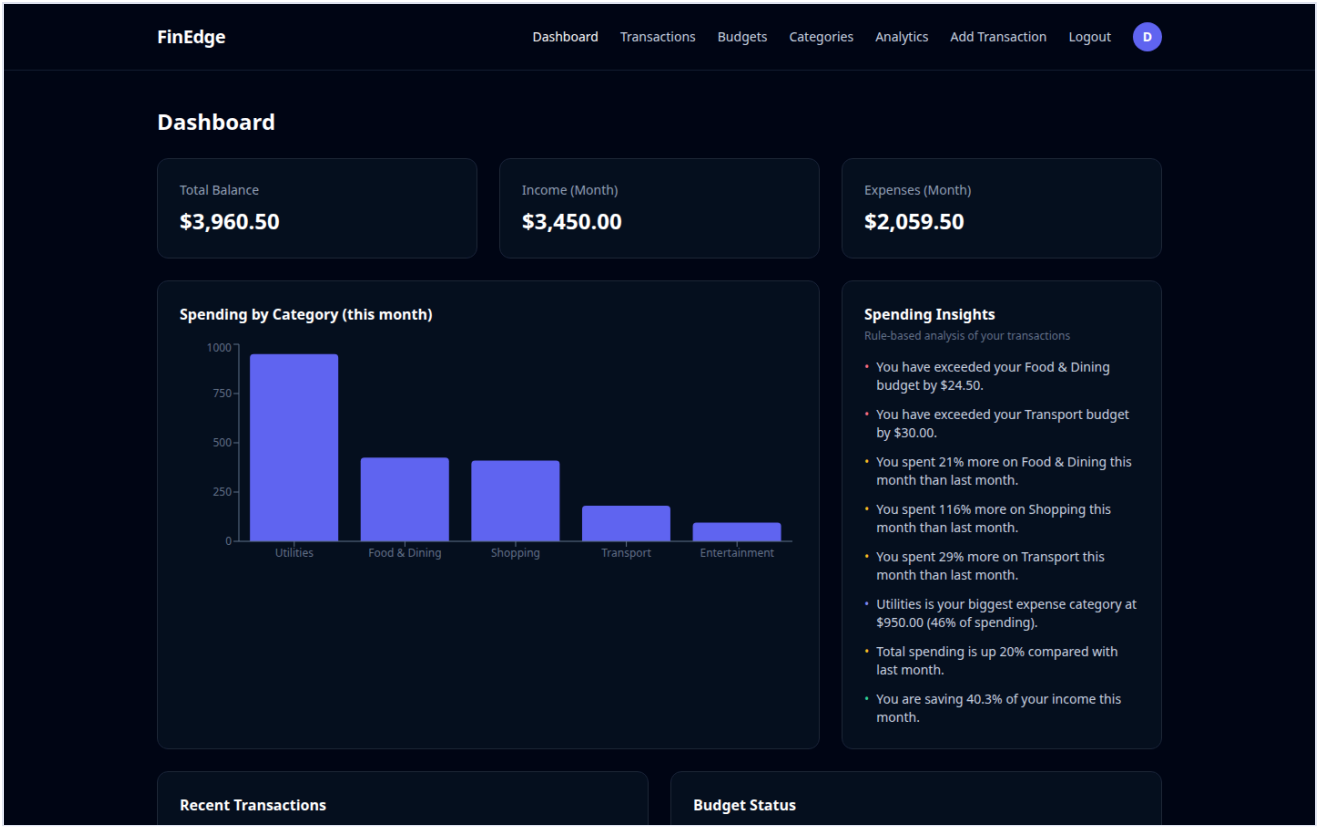


Fig. 39. Rule-based insights for the seeded month. *FR-19, FR-20.*

6b. Developed functionality — mobile (390×844)

Figure 40 — Mobile dashboard



Fig. 40. Dashboard on a phone. *FR-23.*

Figure 41 — Mobile transaction flow

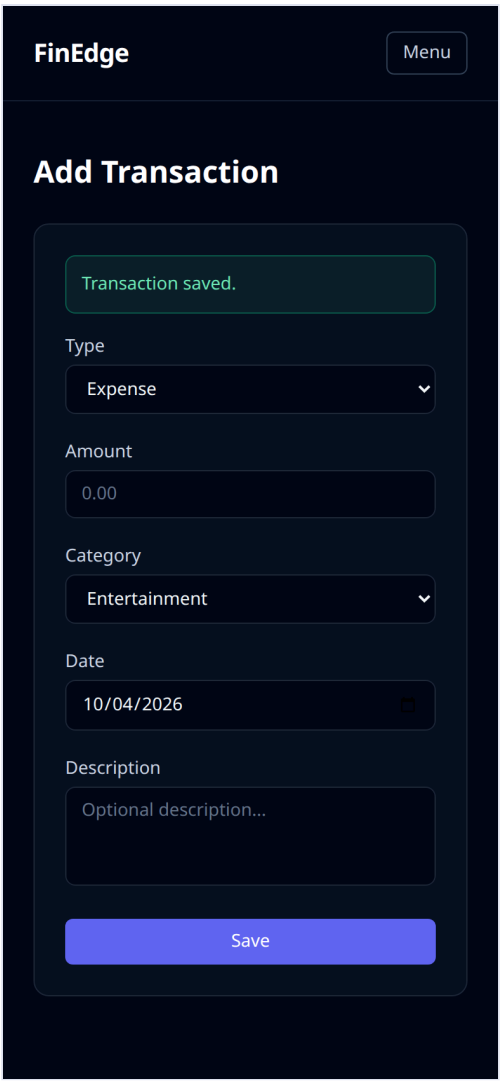


Fig. 41. Add-transaction form after a successful save. *FR-23.*

Figure 42 — Mobile budgets

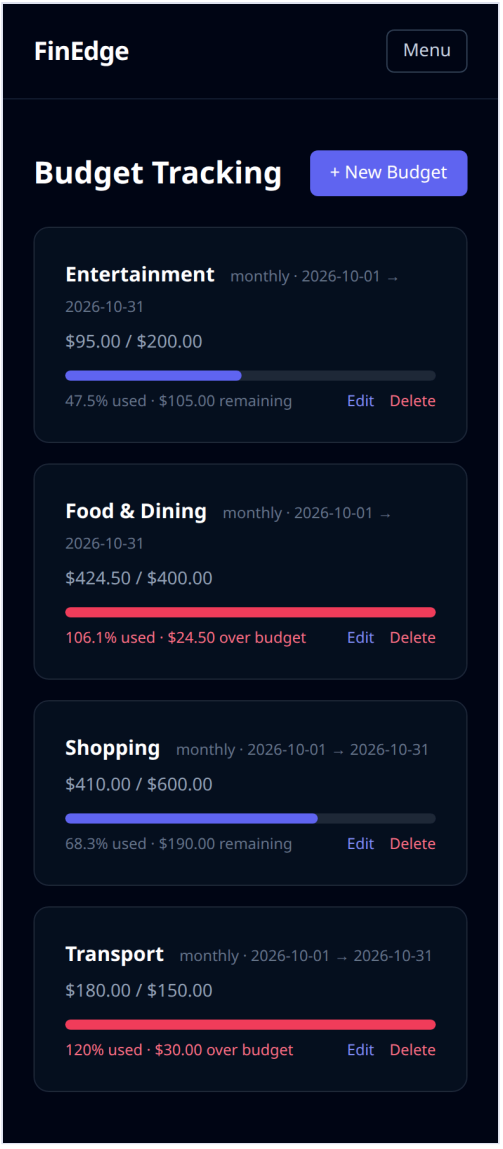


Fig. 42. Budget cards with computed status. *FR-23.*

Figure 43 — Mobile analytics

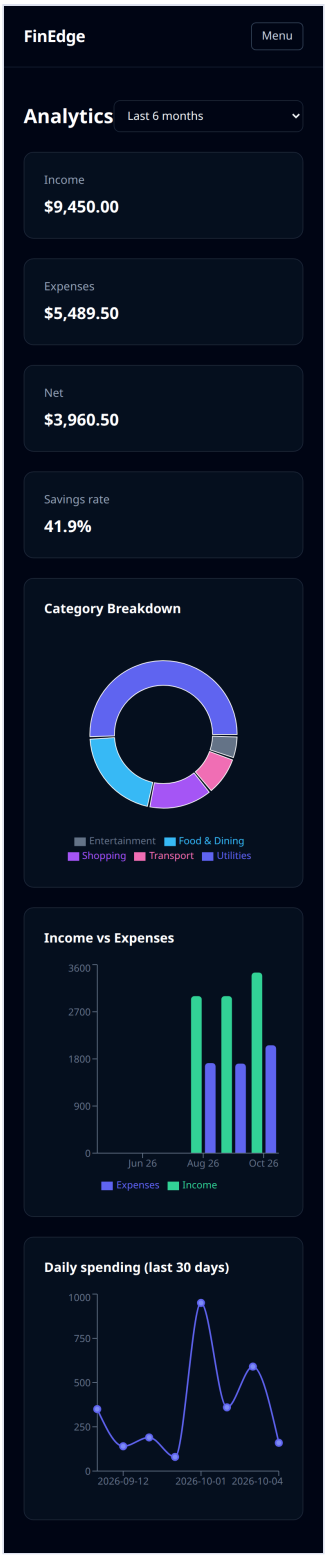


Fig. 43. Charts stacked for a narrow screen. *FR-23.*

7. Tools and technologies

Figure 44 — Declared dependencies

Declared dependencies

```
$ (cd backend && node -p "Object.entries({...require(./package.json).dependencies,...require(./package.json).devDependencies})")
backend dependencies
  bcryptjs ^3.0.3
  cors ^2.8.5
  express ^4.19.2
  jsonwebtoken ^9.0.3
  pg ^8.23.1
  zod ^4.6.5
backend devDependencies
  @eslint/js ^10.0.1
  @stryker-mutator/core ^10.0.0
  @stryker-mutator/vitest-runner ^10.0.0
  @vitest/coverage-v8 ^3.2.7
  eslint ^10.12.0
  globals ^17.13.0
  supertest ^7.3.1
  vitest ^3.2.7
frontend dependencies
  react ^18.3.1
  react-dom ^18.3.1
  react-router-dom ^7.18.2
  recharts ^3.10.1
frontend devDependencies
  @eslint/js ^9.39.5
  @playwright/test ^1.63.0
  @testing-library/jest-dom ^7.0.1
  @testing-library/react ^16.3.3
  @testing-library/user-event ^14.6.7
  @vitejs/plugin-react ^4.3.1
  @vitest/coverage-v8 ^3.2.7
  autoprefixer ^10.4.19
  eslint ^9.39.5
  eslint-plugin-react ^7.37.5
  eslint-plugin-react-hooks ^7.1.1
  globals ^17.13.0
  jsdom ^30.1.1
  postcss ^8.4.38
  tailwindcss ^3.4.4
  vite ^5.3.1
  vitest ^3.2.7
```

Fig. 44. Dependencies and dev tools from backend/package.json and frontend/package.json. *Narrative table: tools-and-technologies.md.*

Tools and technologies (summary)

Area	Technology	Version (lockfile)	Used for
Runtime	Node.js	22	Backend runtime, tooling
Backend	Express	4.x	REST API
	node-postgres (`pg`)	8.x	PostgreSQL client
	bcryptjs	3.x	Password hashing
	jsonwebtoken	9.x	Session tokens
	zod	4.x	Request validation
Database	PostgreSQL	16	Persistent storage; SQL migrations (`backend/src/db/migrations`)
Frontend	React	18	UI
	Vite	5	Dev server and bundler
	Tailwind CSS	3	Styling
	Recharts	3	Charts
	react-router	7	Routing
Unit/integration testing	Vitest	3	Test runner (backend and frontend)
	Supertest	7	HTTP integration tests against the Express app
	React Testing Library, user-event, jest-dom, jsdom	latest	Component tests
	V8 coverage (`@vitest/coverage-v8`)	3	Coverage
Regression / E2E	Playwright	1.63	Real-browser tests at 3 viewports; screenshot capture
Mutation testing	StrykerJS (`@stryker-mutator/core`, `vitest-runner`)	10	Mutation score on business logic
Quality	ESLint (+ react, react-hooks plugins in the frontend)	10 backend / 9 frontend	Static analysis
Containers	Docker, Docker Compose	—	Run frontend, backend and PostgreSQL
CI	GitHub Actions	—	`.github/workflows/ci.yml`
Version control	Git, GitHub	—	GitHub Flow, Conventional Commits, `v1.0.0` tag
Documentation	Markdown, Mermaid	11	SRS, architecture, DFD/ERD/UML sources (`docs/review3/diagrams/*.mmd`)
Report tooling	marked, Playwright (PDF)	—	Builds the report and evidence PDFs from the repository (`docs/review3/tools/`)
Optional	Anthropic Messages API	—	AI-written insights when `AI_API_KEY` is set

Exact versions are pinned in `backend/package-lock.json`, `frontend/package-lock.json` and `docs/review3/tools/package-lock.json`.